International Journal of Forecasting 42 (2026) 776–795

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

New tests of equal forecast accuracy for factor-augmented regressions with weaker loadings

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ARTICLE INFO

Article history:
Dataset link: GitHub - New Tests of Equal Forecast Accuracy for Factor-Augmented Regressions with Weaker Loadings

Keywords:
Forecast accuracy
Factor-augmented regressions
Weak loadings
Principal component analysis (PCA)
Nested models

ABSTRACT

We provide the theoretical foundation for the recent tests of equal forecast accuracy and encompassing by Pitarakis (2023) and Pitarakis (2025), when the competing forecast specification is that of a factor-augmented regression model. This should be of interest to practitioners, as there is no theory justifying the use of these simple and powerful tests in such a context. In pursuit of this, we employ a novel theory to incorporate the empirically well-documented fact of homogeneously/heterogeneously weak factor loadings, and track their effect on the forecast comparison problem.

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1. Introduction

Assessing the *out-of-sample* forecast performance of different models is fundamental for practitioners deciding which specification to employ. Central banks, investment banks, and government economic planning agencies routinely employ forecasting models to make informed decisions and implement effective policies. Therefore, evaluating the *population* predictive ability of various alternative models is for them essential.[1,2] Competing forecasts may differ in setup (e.g. forecasting inflation via linear regression or exponential smoothing), but the focus is often on augmenting a linear model with additional predictors. In such cases, models are nested under the null hypothesis of equal forecast accuracy. Nesting comes naturally in the validation of economic theories. For instance, inflation (π_t) can be forecast using an AR(1) model, or alternatively, using an ARX(1) model as $\pi_t = \gamma_1 + \gamma_2\pi_{t-1} + \gamma_3\iota_{t-1} + \nu_t$, where interest rates ($\iota_t$) serve as exogenous input. Under the null hypothesis then, $\gamma_3 = 0$. Another example of this is forecast accuracy comparisons against random walk models in the field of exchange rates, as considered by Rossi (2005) (see Pitarakis, 2025 for an overview).

While natural, nested comparisons nowadays go hand in hand with a high-dimensional setting. With the availability of large contemporary datasets, researchers and practitioners face an abundance of potential predictors that may or may not improve forecasts beyond standard autoregressive specifications. By far the most popular approach to condense the predictive information is to employ the factor structure (see the survey by Eickmeier & Ziegler, 2008). Here, a large number of potential predictors load on a small number of latent series (factors) that drive their co-movement.[3] Therefore, a factor-augmented forecast is the hallmark example of a nested setup. It

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[1] We stress the *out-of-sample* and *in-population* because 'in-sample' mean-squared-error-based forecasts, and the horse races thereof, are void of statistical grounds; see among others (Diebold, 2015).

[2] The numerical results presented in this manuscript were reproduced by the Editor-in-Chief on 12 November 2025. The results may be sensitive to the Matlab version being used to run the code.

[3] To be precise, factor models are not a dimension-reduction technique per se; it is only the reduced rank assumption on the common component, combined with white noise-like assumptions on the idiosyncratics, which have the desired reduction effect (see e.g. Barigozzi & Hallin, 2024, for an excellent discussion). However, for macroeconomic applications, these are often reasonable assumptions.

https://doi.org/10.1016/j.ijforecast.2025.11.005
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also constitutes a whole research industry with applications in macroeconomics and finance, where principal component analysis (PCA) is the predominant method of factor estimation to extract eigenvectors corresponding to the largest eigenvalues of the sample covariance matrix (see Bai & Ng, 2002, or Bai & Ng, 2006). Hence, it is not surprising to see two parallel strands of econometric literature. The first focuses on the robustness of the principal components procedure itself, while the second develops tests to evaluate out-of-sample forecasts with estimated factors. An important example of the former is empirically well documented and it concerns risks when the loadings through which the factors weight on the high-dimensional set of observables are weak (see the examples in Stock & Watson, 2002b, or Ludvigson & Ng, 2009). Technically, the eigenvalues diverge at a sub-linear rate (see e.g. Uematsu & Yamagata, 2022, and Bai & Ng, 2023). Practically, it limits the informativeness of a dataset and may cloud forecast comparisons. Indeed, the key examples of the second strand are built only on the strong-loadings assumption, and they are the seminal work of Gonçalves et al. (2017) and Stauskas and Westerlund (2022).[4] They derive conditions under which (highly non-standard) asymptotic distributions of the tests of Clark and McCracken (2001) continue to hold when the competing model is augmented with estimated factors. In the current study, we merge both strands. Firstly, we re-establish standard normal inference by borrowing from the recent developments in the theory of nested environments by Pitarakis (2023, 2025). Next, we introduce the possibility of weaker loadings to quantify their effect on the forecast comparison problem.

Statistical comparisons of nested models typically result in a non-standard inference. Particularly, under the null, the population errors of the two competing specifications are identical, leading to zero out-of-sample mean square error (MSE) differentials in the limit, as well as zero asymptotic variances. As a result, the test statistics become asymptotically ill defined. On the contrary, these problems are bypassed in the non-nested case, and asymptotic normality is relatively easy to establish (see Diebold, 2015). Such challenges motivated early works by Clark and McCracken (2001) and McCracken (2007), who introduced adaptive normalizations of the MSEs to recover well-defined asymptotic distributions of the test statistics. However, these distributions are highly non-standard and follow stochastic integrals of Brownian motion that depend on the relative growth rate of in-sample versus out-of-sample observations. Simulation-based approaches for estimating asymptotically valid critical values exist (see Clark & McCracken, 2012, or Hansen & Timmermann, 2015), but their practical implementation is very challenging, which makes the results of Gonçalves et al. (2017) elegant, but impractical.

More recently, Jean-Yves Pitarakis (JYP) proposed a set of tests for predictive accuracy and encompassing in nested models, avoiding the non-standardness of the asymptotic distribution of the statistics under the null, as well as the variance degeneracy issue of existing procedures. Pitarakis (2025) proposes a forecast accuracy test, still based on the MSE comparison across two nested models. However, in comparing MSEs, it uses partially overlapping out-of-sample segments to compute them, rather than the whole, same out-of-sample span. The intuition goes that as long as the fractions of out-of-sample squared forecast errors associated with the two competing specifications are different, the variance of a suitably normalized test statistic involving the MSEs cannot be degenerate. JYP then proves the asymptotic normality of two types of tests under a set of general, nonrestrictive assumptions (more details are provided in Section 2).

Pitarakis (2023) instead offers an encompassing test—meaning that it is based on the forecast encompassing principle (see Hendry & Richard, 1982), for which if one forecast offers no additional value over another, then an optimal convex combination of the two cannot yield a lower squared error loss. Defining $\widetilde{u}_{1,t+1}, \widetilde{u}_{2,t+1}$ as the one-step-ahead forecast errors associated with two alternative forecasts, this principle boils down to testing the population moment restriction: $\mathbb{E}(u_{1,t+1}(u_{1,t+1} - u_{2,t+1})) = 0$. However, suitably normalized sample statistics to test such a restriction are also plagued by the same issues that plague the equal predictive ability tests: variance degeneracy and the nonstandardness of asymptotics. To circumvent these, the simple and brilliant idea of Pitarakis (2023) is to consider the linear combination of two subsample means, in place of a unique sample mean, as additive sample counterparts of the population quantities. Simply enough, if $[1 : k_0]$ is the set of in-sample observations and $[k_0+1 : T]$ the out-of-sample's, this means that the sample counterpart of $\mathbb{E}[u_{1,t+1}u_{2,t+1}]$ is not going to be $(T-k_0)^{-1}\sum_{t=k_0}^{T-1}\widetilde{u}_{1,t+1}\widetilde{u}_{2,t+1}$ but $1/2(m_0^{-1}\sum_{t=k_0}^{m_0+k_0-1}\widetilde{u}_{1,t+1}\widetilde{u}_{2,t+1} + (T - k_0 - m_0)^{-1}\sum_{t=m_0+k_0}^{T-1}\widetilde{u}_{1,t+1}\widetilde{u}_{2,t+1})$, for m_0 being a split point that should be different from $(T - k_0)/2$. The two subgroups of units will have slightly different variances, and that is the key to circumventing the variance degeneracy issue. As for the previous tests, JYP proves asymptotic normality of his proposed encompassing test statistics under a set of general, high-level assumptions.

For their simplicity, robustness, and generality, JYP's tests should be considered the new standard procedures to test for forecast accuracy and encompassing. Crucially, they have the potential to make the out-of-sample evaluation of factor-augmented forecasts much more practical. However, as of now, there is no theory formally justifying their appealing properties in this setting, as his high-level assumptions accommodate only observed, but not estimated, factors. Pitarakis (2023) only considers a factor-augmented data generating process (cf. DGP2) as a robustness check in the simulations.[5] Moreover, given

[4] Their study, however, exploits the common correlated effects estimator by Pesaran (2006). While this method has elegant properties when the data admit a specific structure (e.g. blocks), we focus on PCA, due to its versatility.

[5] Also, upon reviewing his simulations, it appears that the factors are not re-estimated at every roll of the out-of-sample window, but are just estimated once, prior to the recursion, over the whole out-of-sample span. This neglects the very out-of-sample nature of the setting (Pitarakis (2023) simulation scripts are available on the author's GitHub page at https://github.com/jpitarakis/Multi-Step_Encompassing).

the potential for wide applicability, it is only natural to provide the justification by reflecting the risks that occur in practice. Indeed, various routes have been taken to accommodate the weakness of factor loadings, which include local factors (see Freyaldenhoven, 2022) or sparsity, which prescribes a sparse principal component estimator as an alternative (see Uematsu & Yamagata, 2022). Only recently, Bai and Ng (2023) provided a comprehensive theory for the usual principal component estimator to determine the impact of the loading weakness. It is also more general than in the latter studies. Particularly, if the assumptions of Uematsu and Yamagata (2022) were applied to the usual principal components setting, factor loadings would need to be stronger for the factors to be identified.

As the key contribution, we provide the theoretical framework that justifies the use of the whole set of JYP's tests of forecast accuracy and encompassing, when the competing specification is that of a factor-augmented autoregression. The factors are estimated at every roll of the out-of-sample window (recursively), and the factor loadings are allowed to be strong, as well as homogeneously or heterogeneously weak. This leads to a technical contribution, where we re-work the weaker loadings theory in Bai and Ng (2023) to the recursive-estimation setting, which can be of independent interest. We spell out and discuss the assumptions required to show how, in all these settings, JYP's test statistics maintain a Gaussian asymptotic distribution and appealing power properties, as long as the loadings are at most moderately weak. This extra informativeness condition is required to take into account the recursive estimation of factors.

Finally, a few words on the notation we use throughout. Firstly, a is a scalar, $\mathbf{a}$ is a vector, and $\mathbf{A}_t$ is a matrix with t rows. For any generic matrix $\mathbf{A}$, the spectral norm is $\|\mathbf{A}\|_{sp} = \sqrt{\lambda_{\max}(\mathbf{A}'\mathbf{A})}$, while $\|\mathbf{A}\| = \sqrt{\text{tr}(\mathbf{A}'\mathbf{A})}$ is the Frobenius norm with tr(.) being the trace operator. The vectorization of a matrix $\mathbf{A}$ is denoted by vec($\mathbf{A}$), and $\lambda_{\min(\max)}(\mathbf{A})$ denotes the smallest (largest) eigenvalue. Next, $\lfloor x \rfloor$ represents the integer part of x, and M is a positive constant, while $\sup_{a\leq t\leq b}$ ($\inf_{a\leq t\leq b}$) is the supremum (infimum). Moreover, k_0 represents the in-sample observations, while $T-k_0=n$ denotes the out-of-sample observations. Convergence in distribution and probability are given by $\to_d$ and $\to_p$, respectively, while weak convergence is given by $\Rightarrow$. Ultimately, $\widetilde{a}$ and $\widehat{a}$ are quantities estimated under the observed and estimated $\mathbf{f}_t$.

2. Econometric setup & tests

For $t = 1, \ldots, T$, let us consider the following forecasting model:

$$y_{t+1} = \boldsymbol{\theta}'\mathbf{w}_t + \boldsymbol{\beta}'\mathbf{f}_t + u_{t+1} = \boldsymbol{\delta}'\mathbf{z}_t + u_{t+1}, \tag{2.1}$$

where $\mathbf{w}_t \in \mathbb{R}^k$, $\mathbf{f}_t \in \mathbb{R}^r$, which are stacked into $\mathbf{z}_t \in \mathbb{R}^{k+r}$ with conformable parameter vectors: $\boldsymbol{\theta}$, $\boldsymbol{\beta}$, and $\boldsymbol{\delta}$. We consider $\mathbf{w}_t$ the 'known factors', and this can contain both lags of y_t, as well as an intercept, seasonal dummies, or time-period dummies. Instead, $\mathbf{f}_t$ is a vector of 'unknown factors'. In particular, we assume that there exists a panel of N series (which excludes y_t) whose components $x_{i,t}$, $i = 1, \ldots, N$, can be decomposed into two unobservable and mutually orthogonal components: a common component $\chi_{i,t}$ and an idiosyncratic component $e_{i,t}$. Respectively, they represent the co-movements and the individual features of the series. For $\chi_{i,t}$, we assume it to be of low rank, i.e. to be driven linearly by an r-dimensional vector of common static factors $\mathbf{f}_t$, such that $\chi_{i,t} = \boldsymbol{\lambda}_i'\mathbf{f}_t$, with $\boldsymbol{\lambda}_i$ being, for every i, an r dimensional vector of factor loadings. Thus, the decomposition takes the following form:

$$x_{i,t} = \boldsymbol{\lambda}_i'\mathbf{f}_t + e_{i,t}, \quad \text{for } i = 1, \ldots, N \text{ and } t = 1, \ldots, T, \tag{2.2}$$

or, in matrix notation,

$$\mathbf{X} = \mathbf{F}\boldsymbol{\Lambda}' + \mathbf{E}, \tag{2.3}$$

where $\mathbf{F} = (\mathbf{f}_1, \ldots, \mathbf{f}_T)' \in \mathbb{R}^{T\times r}$, $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_N)' \in \mathbb{R}^{N\times r}$ is the matrix of individual factor loadings, and $\mathbf{E} \in \mathbb{R}^{T\times N}$ is the matrix of idiosyncratic components.

We are interested in understanding whether the factor-augmented model ($\boldsymbol{\beta} \neq \mathbf{0}_r$, 'unrestricted') is on average better in terms of out-of-sample forecast accuracy than a simple, possibly autoregressive model ($\boldsymbol{\beta} = \mathbf{0}_r$, 'restricted'). For this purpose, we split the sample T into $T = k_0 + n$, where k_0 and n are the in- and out-of-sample periods, respectively. Conveniently, we let $k_0 = \lfloor T\pi_0 \rfloor$ for $\pi_0 \in (0, 1)$. Then, we produce recursive pseudo-out-of-sample forecasts for the restricted and unrestricted models for $t = k_0, \ldots, T-1$ and compare their errors of the form $\widetilde{u}_{1,t+1} = y_{t+1} - \widetilde{\boldsymbol{\theta}}_t'\mathbf{w}_t$ against $\widetilde{u}_{2,t+1} = y_{t+1} - \widetilde{\boldsymbol{\theta}}'\mathbf{w}_t - \widetilde{\boldsymbol{\beta}}'\mathbf{f}_t = y_{t+1} - \widetilde{\boldsymbol{\delta}}_t'\mathbf{z}_t$ for $t = k_0, \ldots, T-1$. Here and henceforth, the notation $\widetilde{a}$ ($\widehat{a}$) indicates an infeasible (feasible) estimator; as such, the least squares estimator

$$\widetilde{\boldsymbol{\delta}}_t = \left(\sum_{s=1}^{t-1}\mathbf{z}_s\mathbf{z}_s'\right)^{-1}\sum_{s=1}^{t-1}\mathbf{z}_s y_{s+1}, \tag{2.4}$$

is indeed infeasible, given that the factors $\mathbf{f}_t$ are unobserved. However, we are first going to proceed as if $\mathbf{f}_t$ was given. The reason being that the expansions of the feasible test statistics reveal how the key component in this analysis is indeed the difference between the infeasible and feasible forecast error: $\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1}$, similarly to Gonçalves et al. (2017) and Stauskas and Westerlund (2022). The boundedness of functions of this quantity is indeed the key to the main theoretical results in this paper.

2.1. Tests for forecast encompassing and accuracy: observed factors

To begin with, we consider a total of three tests of equal forecast accuracy and encompassing. The test for forecast encompassing is given by Pitarakis (2023) and it has the following form:

$$g_{f,1} = \frac{1}{\widetilde{\omega}_1}\left(\frac{1}{\sqrt{n}}\sum_{t=k_0}^{T-1}\widetilde{u}_{1,t+1}^2 - \frac{1}{2}\left[\frac{n}{m_0}\frac{1}{\sqrt{n}}\sum_{t=k_0}^{k_0+m_0-1}\widetilde{u}_{1,t+1}\widetilde{u}_{2,t+1} + \frac{n}{n-m_0}\frac{1}{\sqrt{n}}\sum_{t=k_0+m_0}^{T-1}\widetilde{u}_{1,t+1}\widetilde{u}_{2,t+1}\right]\right), \tag{2.5}$$

where $m_0 = \lfloor n\mu_0 \rfloor = \lfloor (T - k_0)\mu_0 \rfloor$ is a cut-off point to split the average for $\mu_0 \in (0, 1)$, with $\mu_0 \neq 1/2$; and $\widetilde{\omega}_1$ is the estimated standard deviation of the limiting distribution of the test statistic.

Two further tests of forecast accuracy come from (Pitarakis, 2025):

$$g_{f,2} = \frac{1}{\widetilde{\omega}_2} \frac{n}{l_1^0} \left(\frac{1}{\sqrt{n}} \sum_{t=k_0}^{k_0+l_1^0-1} \widetilde{u}_{1,t+1}^2 - \frac{l_1^0}{l_2^0} \frac{1}{\sqrt{n}} \sum_{t=k_0}^{k_0+l_2^0-1} \widetilde{u}_{2,t+1}^2 \right) \tag{2.6}$$

and

$$g_{f,3} = \frac{1}{\widetilde{\omega}_3} \frac{1}{n(1-\tau_0)} \sum_{l_1=\lfloor n\tau_0 \rfloor+1}^{n} g_{f,2}(l_1, \lfloor n\lambda_2^0 \rfloor). \tag{2.7}$$

Here, $l_j^0 = \lfloor n\lambda_j^0 \rfloor$ for $j = 1, 2$ and $\lambda_j^0 \in (0, 1)$ controls the two portions of the out-of-sample period over which the forecast errors of both models are compared. One can have $l_1^0 > l_2^0$ or vice versa, which means that both portions are overlapping. However, an equality is ruled out in order to avoid the asymptotic degeneracy of variance. This test has a slight disadvantage when compared to that of Pitarakis (2023), because of the overlapping evaluation over the effective sample size. Some data, though in principle minimal data, are lost in the MSE comparison. Note also how (2.7) is simply an average of (2.6) over some chosen feasible set of l_1 for the fixed l_2^0 (or fixed λ_2^0). The tuning parameter $\tau_0 \in (0, 1)$ helps to pick that set.[6] The intuition behind the averaging is the following: if l_2^0 is fixed and l_1 changes, then the MSE of the restricted model accumulates. In effect, the uncertainty over the possible choices of l_1 is integrated.

Pitarakis (2025) makes several suggestions for different avenues of averaging. One that is significant both practically and theoretically is averaging over l_2 while l_1^0 is fixed, such that the MSE of the unrestricted model accumulates. Because in practice the feasible version of the unrestricted model will contain a factor estimation error, we can track whether it interferes with an integration of the uncertainty around l_2. This results in a new statistic:

$$g_{f,4} = \frac{1}{\widetilde{\omega}_4} \frac{1}{n(1-\tau_0)} \sum_{l_2=\lfloor n\tau_0 \rfloor+1}^{n} g_{f,2}(\lfloor n\lambda_1^0 \rfloor, l_2). \tag{2.8}$$

This statistic is not present in Pitarakis (2025), so as an additional contribution, we provide its full analysis. Lastly, $\widetilde{\omega}_j^2$ for $j = 1, \ldots, 4$ represents the variance estimators of the four statistics.

Algorithm 1, below, summarizes $g_{f,1}$ to $g_{f,4}$ and their respective variance estimators $\widetilde{\omega}_1^2, \ldots, \widetilde{\omega}_4^2$. It also provides recommendations for the tuning parameter values, which serve to boost the statistical power of the tests. The recommendations stem from a theoretical and simulation-based investigation by Pitarakis (2025), while in our simulations, we experiment with different values to explore the balance of size and power.

[6] For further discussion on how to choose the tuning parameters, we refer the reader to Pitarakis (2025), Section 3.

Algorithm 1 Operationalizing test statistics.

Select in-sample portion $\pi_0 \in (0, 1)$ and $k_0 = \lfloor T\pi_0 \rfloor$
Obtain out-of-sample: $n = T - k_0$
Initialize with $\widetilde{\boldsymbol{\theta}}_{k_0}$ and $\widetilde{\boldsymbol{\delta}}_{k_0}$ using $s = 1, \ldots k_0$
for $t = k_0 : T - 1$ **do**
  $\widetilde{u}_{1,t+1} = y_{t+1} - \widetilde{\boldsymbol{\theta}}_t' \mathbf{w}_t$, and $\widetilde{u}_{2,t+1} = y_{t+1} - \widetilde{\boldsymbol{\delta}}_t' \mathbf{z}_t$
end for
Estimate ϕ^2 as $\widetilde{\phi}^2 = \frac{1}{n}\sum_{t=k_0}^{T-1} \left(\widetilde{u}_{2,t+1}^2 - \frac{1}{n}\sum_{t=k_0}^{T-1} \widetilde{u}_{2,t+1}^2 \right)^2$
Select *Test* $\in \{g_{f,1}, g_{f,2}, g_{f,3}, g_{f,4}\}$
if *Test* $= g_{f,1}$ **then**
  $\widetilde{\omega}_1^2 = \widetilde{\phi}^2 \frac{(1-2\mu_0)^2}{4\mu_0(1-\mu_0)}$; **recommend**: $\mu_0 \neq 0$; $\mu_0 \to 0.5$;
else if *Test* $= g_{f,2}$ **then**
  $\widetilde{\omega}_2^2 = \widetilde{\phi}^2 \frac{|\lambda_1^0 - \lambda_2^0|}{\lambda_1^0 \lambda_2^0}$; **recommend**: $\lambda_1^0 \neq \lambda_2^0$; λ_1^0, λ_2^0 both close to 1;
else if *Test* $= g_{f,3}$ **then**

$$\widetilde{\omega}_3^2 = \begin{cases} \widetilde{\phi}^2 \frac{(1-\tau_0)^2 + 2\lambda_2^0(1-\tau_0+\ln(\tau_0))}{\lambda_2^0(1-\tau_0)^2} & \text{if } \lambda_2^0 \leq \tau_0, \\ \widetilde{\phi}^2 \frac{1-\tau_0^2 + 2\lambda_2^0((1-\tau_0)\ln(\lambda_2^0) + \tau_0 \ln(\tau_0))}{\lambda_2^0(1-\tau_0)^2} & \text{if } \lambda_2^0 > \tau_0; \end{cases}$$

recommend: $\tau_0 \to 1$; $\lambda_2^0 = 0.5\tau_0 + 0.5$;
else if *Test* $= g_{f,4}$ **then**

$$\widetilde{\omega}_4^2 = \begin{cases} \widetilde{\phi}^2 \frac{(1-\tau_0)^2 + 2\lambda_1^0(1-\tau_0+\ln(\tau_0))}{\lambda_1^0(1-\tau_0)^2} & \text{if } \lambda_1^0 \leq \tau_0, \\ \widetilde{\phi}^2 \frac{1-\tau_0^2 + 2\lambda_1^0((1-\tau_0)\ln(\lambda_1^0) + \tau_0 \ln(\tau_0))}{\lambda_1^0(1-\tau_0)^2} & \text{if } \lambda_1^0 > \tau_0; \end{cases}$$

recommend: $\tau_0 \to 1$; $\lambda_1^0 = 0.5\tau_0 + 0.5$;
end if

Note that for $j = 1, \ldots, 4$, we have $\omega_j^2 = \phi^2 \gamma_j^2$, where γ_j^2 is known. We only need to estimate ϕ^2 with

$$\widetilde{\phi}^2 = \frac{1}{n} \sum_{t=k_0}^{T-1} \left(\widetilde{u}_{2,t+1}^2 - \frac{1}{n} \sum_{t=k_0}^{T-1} \widetilde{u}_{2,t+1}^2 \right)^2, \tag{2.9}$$

where in practice we use the feasible estimator $\widehat{\phi}^2$. Similar sample variance estimators have been widely used in different exercises for factor-augmented regressions (see e.g. Bai & Ng, 2006, Gonçalves et al., 2017, or Yan & Cheng, 2022)

For further exposition, it is convenient to split (2.5)–(2.8) into two components, such that for $j = 1, \ldots, 4$ we have

$$g_{f,j} = \widetilde{\omega}_j^{-1}(g_{f,j,1} + g_{f,j,2}), \tag{2.10}$$

where $g_{f,j,1}$ generates the distribution under the null, and $g_{f,j,2}$ generates power under the alternative. Then, as $T \to \infty$, we have that, under the null, the $\widetilde{\omega}_j^{-1}$-scaled first component converges in distribution to a standard Gaussian density function. To see its key difference from the tests in Clark and McCracken (2001), let $\sigma^2 = \mathbb{V}ar(u_{t+1})$ (unconditional variance). Then, the process $\{u_{t+1}^2 - \sigma^2\}$ generates the distribution, instead of the predictors. For the local power analysis in $g_{f,j,2}$, we impose $\boldsymbol{\beta} = \boldsymbol{\beta}^0 T^{-1/4}$ as the local alternative, similarly to Pitarakis (2023, 2025).

Thus, we require the following assumptions on the forecast error (A.1) and on the factors and predictors (A.2). They are slightly more primitive than in the original studies in order to make them compatible with and sufficient for the feasible setup explored below.

A.1 (Forecast error)

(i) $\{u_t\}$ is a martingale difference sequence with respect to the filtration $\mathcal{F}_{t-1} = \sigma(\mathbf{z}_{t-1}, \dots, \mathbf{X}_{t-1}, \dots, y_{t-1}, \dots)$.
(ii) $\mathbb{E}(u_t^2|\mathcal{F}_{t-1}) = \sigma_t^2$, $\sup_{k_0 \le t \le T-1} \sigma_t^2 = O_p(1)$, $\sigma^2 = \mathbb{E}(\sigma_t^2)$.
(iii) $\sup_{k_0 \le t \le T-1} \mathbb{E}(u_t^4) < M$.
(iv) u_t is independent of all other primitives of the model $\forall i, t, s$.

A.2 (Factors and predictors)

(i) $\sup_{k_0 \le t \le T-1} \mathbb{E}(\|\mathbf{z}_t\|^8) < M$, where $\mathbf{z}_t = (\mathbf{w}_t', \mathbf{f}_t')' \in \mathbb{R}^{k+r}$.
(ii) $\sup_{k_0 \le t \le T-1} \left\| \frac{1}{T} \sum_{s=1}^{t} (\mathbf{z}_s \mathbf{z}_s' - \boldsymbol{\Sigma}_{\mathbf{z}}) \right\| = O_p(T^{-1/2})$ and $\boldsymbol{\Sigma}_{\mathbf{z}} = \mathbb{E}(\mathbf{z}_t \mathbf{z}_t') = \begin{bmatrix} \boldsymbol{\Sigma}_{\mathbf{w}} & \boldsymbol{\Sigma}_{\mathbf{wf}} \\ \boldsymbol{\Sigma}_{\mathbf{wf}}' & \mathbf{I}_r \end{bmatrix}$.
(iii) $\frac{1}{\sqrt{T}} \sum_{s=1}^{t} \mathrm{vec}(\mathbf{z}_s \mathbf{z}_s' - \boldsymbol{\Sigma}_{\mathbf{z}})$ converges weakly to a Brownian motion.

A.1(i) considers the filtration at $t-1$. Hence, this implicitly means that we are looking at the one-step-ahead forecast. As we provide a unified theory for all of JYP's tests, and those in Pitarakis (2025) are only for one step ahead, this choice allows us to treat them all at once. The martingale difference sequence (MDS) assumption is natural in this setting (see the same assumption in Stauskas & Westerlund, 2022 and an equivalent one in Bai & Ng, 2006, Cheng & Hansen, 2015, and Karabiyik & Westerlund, 2021). However, we note that this is merely for convenience, as further steps ahead and serial correlation can be considered. In particular, we comment on how to relax the MDS requirement in Remark 1. As it is likely to occur in practice, we can also allow for conditional heteroskedasticity (e.g. $u_t = \sigma_t \varepsilon_t$, where ε_t is IID(0, 1), and σ_t^2 represents an ARCH/GARCH effect) in the forecast error if (2.9) is replaced by a consistent HAC estimator, since $\{u_{t+1}^2 - \sigma^2\}$ is not MDS anymore. A.1(ii) and A.1(iv) are absent in Gonçalves et al. (2017), but they are fairly standard and allow us to bring down moment requirements on factors and idiosyncratics. Indeed, in the equivalent assumption of A.2 in Gonçalves et al. (2017), 16th moments of factors are required. In our case, it is sufficient to have 8th moments. Generally, parts (i)–(iii) of A.2 ensure that the tests of Pitarakis (2023, 2025) are asymptotically normal as required, but under lower-level conditions. For example, part (iii) is similar to the one in Clark and McCracken (2001) and Stauskas and Westerlund (2022), where $\{\mathrm{vec}(\mathbf{z}_t \mathbf{z}_t' - \boldsymbol{\Sigma}_{\mathbf{z}})\}_{t=k_0}^{T-1}$ follows a mixing sequence of specific size (see Hansen, 1992).

Proposition 1 below gives the said results for $g_{f,j,1}$, $g_{f,j,2}$ under the above assumptions, whereas in Pitarakis (2023, 2025), they are obtained under high-level conditions.

Proposition 1. *Under Assumptions A.1 and A.2, for $j = 1, \dots, 4$ as $T \to \infty$, one has*

$$\widetilde{\omega}_j^{-1} g_{f,j,1} \to_d \mathcal{N}(0, 1).$$

For the local power generating terms $g_{f,j,2}$, for $j = 1, \dots, 4$, one has

$$g_{f,j,2} \to_p \sqrt{1 - \pi_0} \boldsymbol{\beta}^{0\prime} (\mathbf{I}_r - \boldsymbol{\Sigma}_{\mathbf{wf}}' \boldsymbol{\Sigma}_{\mathbf{w}}^{-1} \boldsymbol{\Sigma}_{\mathbf{wf}}) \boldsymbol{\beta}^0.$$

Proof. Online Supplement, Section 2.

Recall the nulls for the encompassing test, $\mathbb{E}(u_{1,t+1}(u_{1,t+1} - u_{2,t+1})) = 0$, and for the forecast accuracy tests, $\mathbb{E}(u_{1,t+1}^2 - u_{2,t+1}^2) = 0$. Both imply that $\boldsymbol{\beta} = \mathbf{0}_r$, and thus that $g_{f,j,2} \to_p 0$. Under the alternatives instead, $g_{f,j,2}$ converges to an expression that explicitly depends on the difference between the diagonal matrix $\boldsymbol{I}_r$ and the quadratic form of the covariances between known and unknown factors with the precision matrix of the known factors: $\boldsymbol{\Sigma}_{\mathbf{wf}}' \boldsymbol{\Sigma}_{\mathbf{w}}^{-1} \boldsymbol{\Sigma}_{\mathbf{wf}}$. Note that by denoting a limit of $g_{f,j,2}$ by ψ, we can give credence to the suggested values in Algorithm 1. For a standard normal CDF $\Phi(.)$ and a quantile related to size α, the power function for each test is $1 - \Phi(q_\alpha - \omega_j^{-1}\psi)$. As $\omega_j^{-1} \propto \gamma_j^{-1}$ for a γ_j that depends on the tuning parameters, we see that the suggestions in Algorithm 1 increase precision and boost the power.

Before we move on to the case of the estimated factors, a word on the further compact notation. We note that the components of (2.5)–(2.8) can be expressed as $\sum_{t=\lfloor f_1(T) \rfloor}^{\lfloor f_2(T) \rfloor} d_u(\widetilde{u}_{1,t+1}, \widetilde{u}_{2,t+1})$ for a loss differential $d_u(.)$, where $\frac{\lfloor f_j(T) \rfloor}{T} \to q_j$ for $j = 1, 2$, such that $q_2 > q_1$. This helps conduct the analysis uniformly over different out-of-sample paths. Therefore, we compactly formulate the results in terms of $\frac{1}{\sqrt{d_T}} \sum_{t=\lfloor f_1(T) \rfloor}^{\lfloor f_2(T) \rfloor} d_u(.)$, where $d_T = (\lfloor f_2(T) \rfloor - \lfloor f_1(T) \rfloor + 1)$. Then, they apply to (2.5)–(2.8) simultaneously.

Example 1. In the case of (2.5), the first component has $\lfloor f_1(T) \rfloor = \lfloor T\pi_0 \rfloor = k_0$ and $\lfloor f_2(T) \rfloor = T - 1$ with $q_1 = \pi_0$ and $q_2 = 1$. The second component has $\lfloor f_1(T) \rfloor = \lfloor T\pi_0 \rfloor = k_0$, $q_1 = \pi_0$, while $\lfloor f_2(T) \rfloor = \lfloor T\pi_0 \rfloor + \lfloor (T - k_0)\mu_0 \rfloor - 1 = k_0 + m_0 - 1$, $q_2 = \pi_0 + (1 - \pi_0)\mu_0$, and so $d_T = m_0$. Then, let $M_T = \sqrt{\frac{m_0}{n}}$, such that we can analyze

$$\frac{1}{\sqrt{n}} \sum_{t=k_0}^{k_0+m_0-1} \widetilde{u}_{1,t+1} \widetilde{u}_{2,t+1} = M_T \times \frac{1}{\sqrt{d_T}} \sum_{t=\lfloor f_1(T) \rfloor}^{\lfloor f_2(T) \rfloor} \widetilde{u}_{1,t+1} \widetilde{u}_{2,t+1},$$

since $M_T = O(1)$. A similar compact expression applies to (2.6), (2.7), and (2.8).

2.2. Tests for forecast encompassing and accuracy: Estimated factors

Clearly, the estimator $\widetilde{\boldsymbol{\delta}}_t$ in (2.4) is unavailable, as the factors are unobserved. Its feasible counterpart is

$$\widehat{\boldsymbol{\delta}}_t = \left(\sum_{s=1}^{t-1} \widehat{\mathbf{z}}_s \widehat{\mathbf{z}}_s' \right)^{-1} \sum_{s=1}^{t-1} \widehat{\mathbf{z}}_s y_{s+1}, \tag{2.11}$$

where $\widehat{\mathbf{z}}_t = (\mathbf{w}_t', \widehat{\mathbf{f}}_t')'$. Therefore, to use (2.11) in the 'for' loop of Algorithm 1, we additionally obtain $\widehat{\mathbf{F}}_t$ by PCA in

the same loop, whose procedure is defined as

$$(\widehat{\boldsymbol{\Lambda}}_t, \widehat{\mathbf{F}}_t) = \underset{\boldsymbol{\Lambda}, \mathbf{F}_t}{\arg\min} \frac{1}{Nt} \sum_{i=1}^{N} \sum_{s=1}^{t} (x_{i,s} - \boldsymbol{\lambda}_i' \mathbf{f}_s)^2 = (t^{-1}\mathbf{X}_t' \widehat{\mathbf{F}}_t, \sqrt{t}\mathbf{U}_{Nt,r}), \tag{2.12}$$

where $\mathbf{U}_{Nt,r}$ is a $t \times r$ matrix of eigenvectors of $(Nt)^{-1}\mathbf{X}_t\mathbf{X}_t'$. For the minimization, the usual required normalizations are that $t^{-1}\widehat{\mathbf{F}}_t'\widehat{\mathbf{F}}_t = \mathbf{I}_r$ and $\widehat{\boldsymbol{\Lambda}}_t'\widehat{\boldsymbol{\Lambda}}_t$ is diagonal, as $\boldsymbol{\Lambda}$ and $\mathbf{F}$ are not identified separately. We stress that $\widehat{\boldsymbol{\Lambda}}_t$ is indexed by t, because we estimate a different loading matrix for each recursion. This implies that the forecast error for the unrestricted model is given by $\widehat{u}_{2,t+1} = y_{t+1} - \widehat{\delta}_t'\widehat{\mathbf{z}}_t = y_{t+1} - \widehat{\boldsymbol{\theta}}_t'\mathbf{w}_t - \widehat{\boldsymbol{\beta}}_t'\widehat{\mathbf{f}}_t$. This estimated forecast error is used in the feasible versions of (2.5)–(2.8).

Example 2. The feasible version of (2.6) is given by

$$g_{\widehat{f},2} = \frac{1}{\widehat{\omega}_2} \frac{n}{l_1^0} \left(\frac{1}{\sqrt{n}} \sum_{t=k_0}^{k_0+l_1^0-1} \widehat{u}_{1,t+1}^2 - \frac{l_1^0}{l_2^0} \frac{1}{\sqrt{n}} \sum_{t=k_0}^{k_0+l_2^0-1} \widehat{u}_{2,t+1}^2 \right), \tag{2.13}$$

where $\widehat{\omega}_2^2$ is the feasible variance estimator.

Our goal is now to show that (2.5)–(2.8) have the same null asymptotic distribution and power properties as in the original references of Pitarakis (2023, 2025). This applies both when $\mathbf{f}_t$ is observed (infeasible setting) and when the factors are estimated (feasible setting) where the loadings are allowed to be strong or weak. This means that the r eigenvalues of the common component covariance matrix can diverge at a sub-linear rate in N, i.e. at rate N^α, for $\alpha \in (0, 1]$. Accommodation of weaker loadings will require specific assumptions, and we follow Bai & Ng, 2023, who require that $N^{-\alpha}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}$ has a positive definite limit. To provide intuition, $\alpha = 1$ gives the usual strong loading case. Within this setting, factors and loadings can be consistently estimated with principal components yielding the usual rate of $O_p(\max(1/\sqrt{N}, 1/\sqrt{T}))$. Instead, $\alpha = 0$ leads to absolutely uninformative loadings, similarly to Onatski (2012). For instance, let $r = 1$, then if $N^{-\alpha}\boldsymbol{\Lambda}'\boldsymbol{\Lambda} = \sum_{i=1}^{N} \lambda_i^2 < \infty$, we have square-summable loadings, which implies that individual loadings are practically zero for highly indexed individuals. All situations in-between give weaker (or 'weakly influential') loadings (see De Mol et al., 2008; Onatski, 2012). This means that as $N \to \infty$ the loadings are too small or too sparse for the corresponding eigenvalues to diverge at rate N (see Barigozzi & Hallin, 2024). Because of this sub-linear divergence, consistent estimation of the common component with PCA is less straightforward. As we also prove below, in the recursive estimation setup, PCA only allows us to recover the factors associated with eigenvalues that diverge at least at a rate equal to N^α, for $\alpha > 1/2$, which is a new result in the PCA literature (see also: Bai & Ng, 2023; Freyaldenhoven, 2022).

Recall that $\widehat{\mathbf{F}}_t$ is the matrix of r eigenvectors corresponding to the r largest eigenvalues of $(Nt)^{-1}\mathbf{X}_t\mathbf{X}_t'$, and $\mathbf{D}_{Nt,r}$ ($\mathbf{D}_{Nt,r}^2$) is a matrix with its singular values (eigenvalues) arranged in decreasing order.[7] Similar decompositions as in Bai and Ng (2023) can therefore be obtained to get an explicit expression of the difference between the estimated factors and a rotated version of the true factors, i.e. $\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}_{Nt,r}'$. This is important, as one never estimates the true factors but always a rotated version of them. Note that for the rotation matrix, we have

$$\mathbf{H}_{Nt,r} = \mathbf{D}_{Nt,r}^{-2}(t^{-1}\widehat{\mathbf{F}}_t'\mathbf{F}_t)(N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}) = \left(\frac{N}{N^\alpha}\mathbf{D}_{Nt,r}^2\right)^{-1}(t^{-1}\widehat{\mathbf{F}}_t'\mathbf{F}_t)(N^{-\alpha}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}), \tag{2.14}$$

which accommodates the fact that we employ the usual principal components procedure when the loadings can be weaker. The next assumption imposes structure on the weaker loadings.

A.3 (Loadings)

(i) $\left\| N^{-\alpha}\boldsymbol{\Lambda}'\boldsymbol{\Lambda} - \boldsymbol{\Sigma}_{\boldsymbol{\Lambda}} \right\| = O_p(N^{-\alpha/2})$ for a positive definite $\boldsymbol{\Sigma}_{\boldsymbol{\Lambda}}$ as $N \to \infty$. The matrix $N^{-\alpha}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}$ is diagonal with distinct eigenvalues ordered in a decreasing fashion. The loadings are independent from $\mathbf{F}_t$ for all t.

(ii) $\mathbb{E}\left[\left(\frac{1}{N^{\alpha/2}}\sum_{i=1}^{N}\mathbf{f}_s'\boldsymbol{\lambda}_i e_{i,t}\right)^2\right] < M$ and $\left\|\frac{1}{\sqrt{N^\alpha T}}\mathbf{F}_t'\mathbf{E}_t\boldsymbol{\Lambda}\right\| = O_p(1)$, $\forall t, s$.

(iii) $\mathbb{E}\left(\left\|\frac{1}{N^{\alpha/2}}\sum_{i=1}^{N}\boldsymbol{\lambda}_i' e_{i,t}\right\|^8\right) < M$, $\forall t$.

(iv) $\left\|\frac{1}{N^\alpha T}\mathbf{e}_i'\mathbf{E}_t\boldsymbol{\Lambda}\right\| = O_p(N^{-\alpha}) + O_p(N^{-\alpha/2}T^{-1/2})$, $\forall i = 1, \ldots, N$ and $\forall t$.

(v) $\sup_{k_0 \le t \le T-1}\|T^{-1/2}\sum_{s=1}^{t}\mathbf{f}_s e_{i,s}\|^2 \le M$, $\forall i = 1, \ldots, N$, $\left\|(NT)^{-1}\mathbf{e}_s'\mathbf{E}_t'\mathbf{F}_t\right\| = O_p(N^{-1}) + O_p(T^{-1})$, $\forall t, s$.

We treat the loadings as random and make them independent from $\mathbf{F}_t$ in order to simplify some arguments. Alternatively, we can impose $\mathbb{E}(\mathbf{f}_t\mathbf{f}_t'|\boldsymbol{\Lambda}) = \mathbf{I}_r$. They can also be treated as fixed, similarly to Gonçalves et al. (2017). Moreover, as in the latter study, we impose a convergence rate to facilitate the quantification of our analysis (under $\alpha = 1$, it coincides with the natural $N^{-1/2}$ rate) in the recursive setup. Naturally, this slightly strengthens the original assumption of Bai and Ng (2023). In general, (i) is a central assumption that allows us to conduct an asymptotic analysis uniformly in loading strength. A similar formulation is used by Uematsu and Yamagata (2022), but in the context of the sparse principal component estimator, and by He et al. (2025), who exploit the minimization of the Huber loss function to estimate the loadings. However, neither of these studies deals with hypothesis testing in a forecasting setting. Recently, Boot and Keijsers (2025) used the assumptions of Bai and Ng (2023) to compare

[7] Let us note that the singular value decomposition (and the subsequent decomposition) is here conducted on the $t \times t$ covariance for $\mathbf{X}'$ i.e. $\frac{1}{Nt}\mathbf{X}\mathbf{X}'$, following the work of Bai and Ng (2023). The same decomposition and asymptotic expansions below can be done in the (more traditional) case of taking eigenvectors of the $N \times N$ covariance $\frac{1}{Nt}\mathbf{X}'\mathbf{X}$ (see Barigozzi et al., 2024; Stock & Watson, 2002a). Both singular value decompositions return the same set of singular values and nothing changes as long as the interest is in modeling static principal eigenvectors.

principal components forecasts with ridge regression and random projections. The remaining assumptions are the higher-level conditions from both Gonçalves et al. (2017) (e.g. iii) and Bai and Ng (2023) (iv and v). The latter are also utilized by Boot and Keijsers (2025), but they would be directly implied if $\mathbf{f}_t$ and $e_{i,s}$ were independent for all i, s, t. The equivalent of (iii) in Gonçalves et al. (2017) requires 16th moments, whereas we deem it very strong and bring it down due to A.1(iv).

2.2.1. Homogeneous weak loadings

We obtain an expansion of $\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}'_{Nt,r}$ by implementing further modifications from both Bai and Ng (2023) and Gonçalves et al. (2017). We start with the case of homogeneous loadings and move in the next section to the case of heterogeneous loadings, i.e. when the loadings are allowed to be weaker to different degrees. We define the following scalar quantities:

$$\gamma_{l,s,\alpha} = \frac{1}{N^\alpha}\sum_{i=1}^{N}\mathbb{E}(e_{i,l}e_{i,s}) = \frac{N}{N^\alpha}\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}(e_{i,l}e_{i,s}) = \frac{N}{N^\alpha}\gamma_{l,s}, \tag{2.15}$$

$$\xi_{l,s,\alpha} = \frac{N}{N^\alpha}\frac{1}{N}\sum_{i=1}^{N}(e_{i,l}e_{i,s} - \mathbb{E}(e_{i,l}e_{i,s})) = \frac{N}{N^\alpha}\xi_{l,s}, \tag{2.16}$$

$$\eta_{l,s,\alpha} = \frac{1}{N^\alpha}\sum_{i=1}^{N}\mathbf{f}'_l\boldsymbol{\lambda}_i e_{i,s}, \quad \nu_{l,s,\alpha} = \frac{1}{N^\alpha}\sum_{i=1}^{N}\mathbf{f}'_s\boldsymbol{\lambda}_i e_{i,l}. \tag{2.17}$$

Note that only the terms that depend on $\boldsymbol{\lambda}_i$ are directly scaled by $N^{-\alpha}$. Since (2.15) and (2.16) are functions of the idiosyncratics only, N/N^α can be seen as a 'penalty' term on the overall rate, as we estimate potentially weaker loadings with principal components. Therefore, we obtain the following decomposition for a row s:

$$\widehat{\mathbf{f}}_s - \mathbf{H}_{Nt,r}\mathbf{f}_s = \left(\frac{N}{N^\alpha}\mathbf{D}^2_{Nt,r}\right)^{-1}\left(\frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\gamma_{l,s,\alpha} + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\xi_{l,s,\alpha} + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\eta_{l,s,\alpha} + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\nu_{l,s,\alpha}\right). \tag{2.18}$$

This is an important term, appearing when expanding $\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1}$, that is, the difference between the infeasible and feasible estimated forecast errors. Specifically, let $\boldsymbol{\Phi}_{Nt,r} = \mathrm{diag}(\mathbf{I}_k, \mathbf{H}_{Nt,r}) \in \mathbb{R}^{(k+r)\times(k+r)}$, where $\mathbf{H}_{Nt,r}$ is the rotation matrix defined in (2.14). Then, in line with Gonçalves et al. (2017) and Stauskas and Westerlund (2022), we obtain

$$\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1} = (\widehat{\mathbf{z}}_t - \boldsymbol{\Phi}_{Nt,r}\mathbf{z}_t)'(\boldsymbol{\Phi}^{-1}_{Nt,r})'(\widetilde{\boldsymbol{\delta}}_t - \boldsymbol{\delta}) + \widehat{\mathbf{z}}'_t(\widetilde{\boldsymbol{\delta}}_t - (\boldsymbol{\Phi}^{-1}_{Nt,r})'\widetilde{\boldsymbol{\delta}}_t) + (\widehat{\mathbf{z}}_t - \boldsymbol{\Phi}_{Nt,r}\mathbf{z}_t)'(\boldsymbol{\Phi}^{-1}_{Nt,r})'\boldsymbol{\delta} = I + II + III. \tag{2.19}$$

Because it is a scalar, we have that $III = (\widehat{\mathbf{f}}_t - \mathbf{H}_{Nt,r}\mathbf{f}_t)'(\mathbf{H}^{-1}_{Nt,r})'\boldsymbol{\beta}$, and III is therefore absent when $\boldsymbol{\beta} = \mathbf{0}_r$ (under the null). The component I, instead, reveals that the asymptotic equivalence is ensured when the factors are estimated consistently and the infeasible forecasting model is well-specified ($\|\widetilde{\boldsymbol{\delta}}_t - \boldsymbol{\delta}\| = o_p(1)$). Term II additionally requires that the feasible OLS estimator of the parameters is asymptotically equivalent to the infeasible one. Because $\widehat{\boldsymbol{\delta}}_t$ employs the factors which are only identified up to a rotation, the former is naturally rotated as well.[8]

The expression (2.19) is the integral part of the feasible versions of (2.5)–(2.8). To demonstrate their asymptotic equivalence, it is useful to introduce the following quantities:

$$A = \frac{1}{\sqrt{d_T}}\sum_{t=\lfloor f_1(T)\rfloor}^{\lfloor f_2(T)\rfloor}(\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1})^2, \tag{2.20}$$

$$B = \frac{1}{\sqrt{d_T}}\sum_{t=\lfloor f_1(T)\rfloor}^{\lfloor f_2(T)\rfloor}\widetilde{u}_{1,t+1}(\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1}), \tag{2.21}$$

$$C = \frac{1}{\sqrt{d_T}}\sum_{t=\lfloor f_1(T)\rfloor}^{\lfloor f_2(T)\rfloor}\widetilde{u}_{2,t+1}(\widetilde{u}_{2,t+1} - \widehat{u}_{2,t+1}), \tag{2.22}$$

$$D = |\widetilde{\omega}_j^2 - \widehat{\omega}_j^2| \quad \text{for} \quad j = 1, \ldots, 4, \tag{2.23}$$

which are expressed in compact notation in order to be applied to every statistic under consideration. Then, for $j = 1, \ldots, 4$,

$$g_{\widehat{f},j} = g_{f,j} + g_{f,j}\left(\frac{\widetilde{\omega}_j}{\widehat{\omega}_j} - 1\right) + q_j(A, B, C, D), \tag{2.24}$$

where $q_j(.)$ is a function, such that $|q_j(A, B, C, D)| = o_p(1)$ if A, B, and C are negligible and $\widehat{\omega}_j = \widetilde{\omega}_j + o_p(1)$. Therefore, the asymptotic equivalence holds for $j = 1, \ldots, 4$ if (2.20)–(2.23) are negligible. The expansion of all the infeasible statistics can be found in the Online Supplement.

Example 3. Let us examine the feasible versions of (2.7) and (2.8):

$$g_{\widehat{f},3} = \frac{1}{\widehat{\omega}_3}\frac{1}{n(1-\tau_0)}\sum_{l_1=\lfloor n\tau_0\rfloor+1}^{n} g_{\widehat{f},2}(\lfloor n\lambda_2^0\rfloor, l_1),$$

$$g_{\widehat{f},4} = \frac{1}{\widehat{\omega}_4}\frac{1}{n(1-\tau_0)}\sum_{l_2=\lfloor n\tau_0\rfloor+1}^{n} g_{\widehat{f},2}(\lfloor n\lambda_1^0\rfloor, l_2).$$

Then, their expansions in the spirit of (2.24) are given by Box I

Contrary to $j = 3$, the factor estimation error accumulates over the choices of l_2 under $j = 4$. Clearly, in both cases for $|q_j(A, C, D)| = o_p(1)$, it is sufficient to have $|A| = o_p(1)$ and $|C| = o_p(1)$. Additionally, if $|\widetilde{\omega}_j^2 - \widehat{\omega}_j^2| = o_p(1)$, then $g_{\widehat{f},j} = g_{f,j} + o_p(1)$ for $j = 3, 4$.

The structure of (2.20)–(2.22) reveals how the behavior of the out-of-sample average of (2.19) dictates the overall asymptotic analysis. For this, we state more assumptions regarding the idiosyncratic components.

A.4 (Idiosyncratics)

(i) $\mathbb{E}(\mathbf{E}_t|\mathbf{F}_t, \boldsymbol{\Lambda}) = \mathbf{0}_{t\times N}$.
(ii) $\sup_{k_0\le t\le T-1}\mathbb{E}(|e_{i,t}|^4) < M$, $\forall i, t$.

[8] Indeed, by the Frisch–Waugh–Lovell argument, the second component is $(\mathbf{H}^{-1}_{Nt,r})'\widetilde{\boldsymbol{\alpha}}_t = (\mathbf{H}'_{Nt,r}\mathbf{F}'_t\mathbf{M}_{\mathbf{W}}\mathbf{F}_t\mathbf{H}_{Nt,r})^{-1}\mathbf{H}'_{Nt,r}\mathbf{F}'_t\mathbf{M}_{\mathbf{W}}\mathbf{y}_t$, where $\mathbf{M}_{\mathbf{W}}$ is the projection matrix onto the orthogonal complement of the observed predictors.

$$
g_{\widehat{f},j} = \begin{cases} (j=3)\ \ g_{f,3} + g_{f,3}\left(\frac{\widetilde{\omega}_3}{\widehat{\omega}_3} - 1\right) \\ + \underbrace{\frac{1}{\widehat{\omega}_3}\frac{n}{l_2^0}\left(\frac{n-\lfloor n\tau_0\rfloor}{n(1-\tau_0)}\right)\left(\frac{2}{\sqrt{n}}\sum_{t=k_0}^{k_0+l_2^0-1}\widetilde{u}_{2,t+1}(\widetilde{u}_{2,t+1}-\widehat{u}_{2,t+1}) - \frac{1}{\sqrt{n}}\sum_{t=k_0}^{k_0+l_2^0-1}(\widetilde{u}_{2,t+1}-\widehat{u}_{2,t+1})^2\right)}_{q_3(A,C,D)}, \\ (j=4)\ \ g_{f,4} + g_{f,4}\left(\frac{\widetilde{\omega}_4}{\widehat{\omega}_4} - 1\right) \\ + \underbrace{\frac{1}{n(1-\tau_0)}\frac{1}{\widehat{\omega}_4}\sum_{l_2=\lfloor n\tau_0\rfloor}^{n}\frac{n}{l_2}\left(\frac{2}{\sqrt{n}}\sum_{t=k_0}^{k_0+l_2-1}\widetilde{u}_{2,t+1}(\widetilde{u}_{2,t+1}-\widehat{u}_{2,t+1}) - \frac{1}{\widehat{\omega}_4}\frac{1}{\sqrt{n}}\sum_{t=k_0}^{k_0+l_2-1}(\widetilde{u}_{2,t+1}-\widehat{u}_{2,t+1})^2\right)}_{q_4(A,C,D)}. \end{cases}
$$

Box I.

(iii) $\mathbb{E}\left[\left(\frac{1}{\sqrt{N}}\sum_{i=1}^{N}(e_{i,t}e_{i,s} - \mathbb{E}(e_{i,t}e_{i,s}))\right)^4\right] < M.$

(iv) $\frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}[\mathbb{E}(e_{i,t}e_{i,s})]^2 < M.$

(v) $\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{1}{T}\sum_{s_1=1}^{T}\sum_{s_2=1}^{T}\left|Cov\left(e_{i,s_1}e_{j,s_1}, e_{i,s_2}e_{j,s_2}\right)\right| = O_p(1).$

(vi) Let $\mathbb{E}(\mathbf{e}_t\mathbf{e}_t') = \boldsymbol{\Sigma}_e \in \mathbb{R}^{N\times N}$. There exists $m > 0$ and some positive M, such that $\lambda_{\min}(\boldsymbol{\Sigma}_e) > m$ and $\|\boldsymbol{\Sigma}_e\|_{sp} < M$.

(vii) $\sup_{k_0\le t\le T-1}\|\mathbf{E}_t\|_{sp} = \max\{\sqrt{N}, \sqrt{T}\}$.

A.4 is similar to the conditions applied by Gonçalves et al. (2017), except for the lower moment requirement. This allows the idiosyncratics to be weakly dependent over time and cross-sectionally. Part (vii) is the same as in Bai and Ng (2023), but it is required to hold uniformly in t. We formulate it in terms of T, because $t = \lfloor sT\rfloor$ for some $s \in (0, 1)$, due to the recursive setup. It is primarily used to simplify proofs when the loading weakness is heterogeneous, but it also helps to improve convergence rates.

Lemma 1 below formalizes the asymptotic behavior of the out-of-sample average of (2.19) by providing the rate of the average square factor approximation error over the recursive samples, when the loadings are weaker. In addition, it provides the uniform rate of the in-sample factor approximation error.

Lemma 1. *Under A.1–A.4, as* $(N, T) \to \infty$,

(i.) $\frac{1}{d_T}\sum_{t=\lfloor f_1(T)\rfloor}^{\lfloor f_2(T)\rfloor}\left\|\widehat{\mathbf{f}}_t - \mathbf{H}_{Nt,r}\mathbf{f}_t\right\|^2 = O_p\left(\frac{N^2}{N^{2\alpha}}\frac{1}{T}\right) + O_p(N^{1-2\alpha}),$

(ii.) $\sup_{k_0\le t\le T-1}\frac{1}{t}\left\|\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}_{Nt,r}'\right\|^2 = O_p\left(\frac{N^2}{N^{2\alpha}}\frac{1}{k_0}\right) + O_p(N^{1-2\alpha}).$

Proof. Online Supplement, Section 3.2.

Lemma 1 can be seen as an extension of Theorem 4.1 in Gonçalves et al. (2017) to the case of weaker loadings and different alternative out-of-sample paths. If $\alpha = 1$ (strong loadings), we return to the usual rate of $O_p(\max\{1/N, 1/T\})$, which coincides with the result in the latter study. Note that the result immediately implies that we must have $\frac{N}{N^\alpha}\frac{1}{\sqrt{T}} \to 0$ and $\alpha > 0.5$ to consistently estimate the factor space in both in- and out-of-sample cases. In contrast, the theory in Bai and Ng (2023) requires a lower bound of α different from 0 only for inference exercises, but not consistency. The difference arises because in part (i), we consider a recursive setup where the rotation matrix in (2.14) changes for every $t = k_0, \dots, T-1$, and therefore this extra informativeness condition needs to be satisfied. Part (ii) of Lemma 1 deals with the average square in-sample factor estimation error, where we average over $s = 1, \dots, t$ as $\frac{1}{t}\left\|\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}_{Nt,r}'\right\|^2 = \frac{1}{t}\sum_{s=1}^{t}\left\|\widehat{\mathbf{f}}_s - \mathbf{H}_{Nt,r}\mathbf{f}_s\right\|^2$. We use $\widehat{\mathbf{F}}_{t-1}$ when obtaining the feasible $\widehat{\boldsymbol{\delta}}_t$ for each recursion. Interestingly, $\alpha > 0.5$ is sufficient but not necessary for part (ii) to hold, since we average for a given $\mathbf{H}_{Nt,r}$. Hence, the approximation rate can be improved with higher-level conditions in both homogeneous and heterogeneous cases, as we point out in Remark 2. However, part (i) is responsible for the general out-of-sample approximation and determines the behavior of $g_{\widehat{f},j}$ for $j = 1, \dots, 4$.

The following result employs part (ii) of Lemma 1 to demonstrate the uniform equivalence of feasible and infeasible OLS estimators.

Lemma 2. *Under Assumptions A.1–A.4, as* $(N, T) \to \infty$ *we have*

$$\sup_{k_0\le t\le T-1}\left\|T^{1/4}(\widehat{\boldsymbol{\delta}}_t - (\boldsymbol{\Phi}_{Nt,r}^{-1})'\widetilde{\boldsymbol{\delta}}_t)\right\| = o_p(1)$$

Proof. Online Supplement, Section 4.2.

Similarly to Lemmas 1 and 2 can be seen as a generalization of Lemma 4.1 of Gonçalves et al. (2017) to weaker loadings. This follows from the direct application of our Lemma 1(ii) when establishing the uniform consistency of $\widehat{\boldsymbol{\delta}}_t$ for (the rotated) $\widetilde{\boldsymbol{\delta}}_t$. Hence, $\alpha > 0.5$ plays a role as well. The implication of the lemma is that the rate of consistency is $o_p(T^{-1/4})$, unlike in the latter study, where it is $o_p(T^{-1/2})$. The difference arises through two related channels. Firstly, the tests of Pitarakis (2023, 2025) have different local power properties. Indeed, we specify

$\boldsymbol{\beta} = \boldsymbol{\beta}^0 T^{-1/4}$, while the tests of Clark and McCracken (2001) explored by Gonçalves et al. (2017) require $\boldsymbol{\beta} = \boldsymbol{\beta}^0 T^{-1/2}$. Secondly, we chose a different proving technique, since we only needed to demonstrate consistency, but the exact rate is less important, due to the different asymptotic properties of our tests.

In order to move on to the main results, we introduce the last assumption.

A.5 (N, T **expansion rates**)
(i) $\frac{N}{N^\alpha}\frac{1}{T^{1/4}} \to c > 0$, as $(N,T) \to \infty$, (ii) $\sqrt{T}N^{-\alpha} \to 0$, for $\alpha \in (0.5, 1)$.

A.5(i) should be seen as a device to make convergence rates more transparent when the loadings are indeed weaker. Specifically, to prove that B and C in (2.21) and (2.22) are negligible, we need $\frac{N}{N^\alpha}\frac{1}{\sqrt{T}} = o(1)$ as before, but at the same time $T^{1/4}\frac{N}{N^\alpha}\frac{1}{\sqrt{T}} = O(1)$. Thus, under A.4(i), $\frac{N}{N^\alpha}\frac{1}{\sqrt{T}} = O_p(T^{-1/4})$, which is the rate that can be incorporated in the subsequent analysis. Requirements in the spirit of A.5(ii) are often met in the PCA literature, and it helps to ensure that the factor estimation error does not accumulate too fast with the expansion of N and T. Here, it is identical to the assumption needed for inference in Lemma 4 of Bai and Ng (2023). Under $\alpha = 1$ (strong loadings), this naturally coincides with the requirement of $\sqrt{T}N^{-1} = o(1)$ in Gonçalves and Perron (2014) or Gonçalves et al. (2017). Both here and in the latter study, this requirement targets sums over an MDS process $\{(\widehat{\mathbf{f}}_t - \mathbf{H}_{Nt,r}\mathbf{f}_t)u_{t+1}\}_{t=k_0}^{T-1}$ and makes sure that they remain asymptotically negligible. Such terms appear by applying (2.19) to components B and C in (2.21) and (2.22), respectively.

Lemma 3, below, is the central outcome that utilizes the interim results discussed above.

Lemma 3. *Under Assumptions A.1–A.5, as* $(N,T) \to \infty$, *we have that A–D are asymptotically negligible.*

Proof. Online Supplement, Section 4.2.

To our knowledge, this is the first result that controls the factor estimation error in the out-of-sample context uniformly in loading strength and alternative out-of-sample paths. Apart from the desired results on A, B and C, we can also see that the feasible variance estimator is asymptotically equivalent to the infeasible one, because

$$\widehat{\phi}^2 = \frac{1}{n}\sum_{t=k_0}^{T-1}\left(\widehat{u}_{2,t+1}^2 - \frac{1}{n}\sum_{t=k_0}^{T-1}\widehat{u}_{2,t+1}^2\right)^2 = \widetilde{\phi}^2 + o_p(1), \tag{2.25}$$

under our assumptions. Note that A - D remain negligible under conditional heteroskedasticity. However, we leave the full analysis of HAC version of D for future research. This provides the last missing piece to establish the equivalence result.

Theorem 1. *Under Assumptions A.1–A.5, as* $(N,T) \to \infty$, *we have*

$$g_{\widehat{f},j} = g_{f,j} + o_p(1), \quad \text{for } j = 1, \dots, 4$$

Proof. The proof follows from the application of Lemma 3.

The main message of Theorem 1 is not only that we are able to use the battery of new statistics (2.5)–(2.8) in the popular context of factor-augmented forecasts, but also that they are robust to weaker factor loadings, as long as $\alpha > 0.5$. This result is a companion to Theorem 4.2 in Gonçalves et al. (2017). While it does not necessarily nest their results, due to different statistics and their local power properties, we have the same approximation rates under strong loadings ($\alpha = 1$).

Before we generalize our results to the heterogeneously weak loadings, it is important to illustrate how Theorem 1 can be used to improve the statistics. The tests of equal forecast accuracy in (2.6) and (2.7) depend on the tuning parameters λ_1^0 and λ_2^0. Naturally, while suggestions on their values are provided in Algorithm 1, your choice still alters the power properties of the tests, as argued by Pitarakis (2025). To bypass this issue, power-enhanced versions of the statistics are presented, e.g.

$$g_{f,2}^{adj} = g_{f,2} + \widetilde{\zeta}(\lambda_1^0, \lambda_2^0),$$

where $\widetilde{\zeta}(\lambda_1^0, \lambda_2^0) = \frac{1}{\widetilde{\omega}_2}\frac{1}{\lambda_2^0}\frac{1}{\sqrt{n}}\sum_{t=k_0}^{k_0+l_2^0-1}(\widetilde{u}_{1,t+1} - \widetilde{u}_{2,t+1})^2$ in the spirit of Fan et al. (2015). Clearly, the power-adjustment term is infeasible and we must replace it with $\widehat{\zeta}(\lambda_1^0, \lambda_2^0)$. Proposition 2 below demonstrates that the power enhancement procedures remain valid.

Proposition 2. *Under Assumptions A.1–A.5, as* $(N,T) \to \infty$, *we have*

$$g_{\widehat{f},2}^{adj} = g_{\widehat{f},2} + \widehat{\zeta}(\lambda_1^0, \lambda_2^0) = g_{f,2} + \widetilde{\zeta}(\lambda_1^0, \lambda_2^0) + o_p(1),$$

and the same holds for $g_{\widehat{f},3}^{adj}$ *and* $g_{\widehat{f},4}^{adj}$, *where the adjustment term is an appropriate average of* $\widehat{\zeta}(\lambda_1^0, \lambda_2^0)$.

Proof. Online Supplement, Section 4.4.

The adjustment term $g_{\widehat{f},4}^{adj}$ concerns our new statistic $g_{\widehat{f},4}$ in (2.8). We relegate its asymptotic analysis together with a broader discussion on how the power adjustment terms are constructed to Section 4.4 of the Online Supplement.

Remark 1. Assumption A.1 implies that the forecast errors are uncorrelated, and so the model is dynamically correctly specified. In practice, we may encounter measurement errors, for example, which force $\{u_t\}$ to be correlated over time. Moreover, if we consider h-step-ahead forecasts, u_{t+h} typically follows a moving average process of order $h-1$ (see Assumption R in Cheng & Hansen, 2015). To account for such possibilities for any $h \geq 1$, we can introduce three high-level conditions instead of Assumption A.1(i) that go beyond moving average processes:

(i) $\frac{1}{\sqrt{T}}\sum_{j=1}^{\lfloor s(T-h)\rfloor}(u_{j+h}^2 - \sigma^2) \Rightarrow \phi B(s)$ as $T \to \infty$, where $B(s)$ is a standard Brownian motion on $s \in [0,1]$, and ϕ^2 is now the long-run variance.
(ii) $\sup_{q\in(0,1)}\left\|\frac{1}{\sqrt{T}}\sum_{s=1}^{\lfloor q(T-h)\rfloor}\mathbf{z}_s u_{s+h}\right\| = O_p(1)$.

(iii) $\left\| \frac{1}{\sqrt{T}} \sum_{t=k_0}^{T-h} \left(\mathbf{A}_{\lfloor qT \rfloor} \frac{1}{\sqrt{T}} \sum_{s=1}^{t-h} \mathbf{z}_s u_{s+h} \right) \mathbf{z}_t u_{t+h} \right\| = O_p(1)$, for $\mathbf{A}_{\lfloor qT \rfloor}$ independent from u_t for all t and such that $\mathbf{A}_{\lfloor qT \rfloor} \to_p \mathbf{A}$ positive definite uniformly.

Conditions (i) and (ii) often appear in the literature when the forecast errors are dependent over time (see e.g. Pitarakis, 2025). For instance, they can be implied if $\{\mathbf{z}_s u_{s+h}\}_{s=1}^{t-h}$ is a mixing of a relevant size. Part (*iii*) mainly concerns the term analyzed in Lemma 2. Note that part (iii) resembles an object that converges weakly to a stochastic integral (see Hansen, 1992). Similar terms are uncovered when sums over an MDS process $\{(\widehat{\boldsymbol{\delta}}_t - (\boldsymbol{\Phi}_{Nt,r}^{-1})'\widetilde{\boldsymbol{\delta}}_t)'\mathbf{z}_t u_{t+1}\}_{t=k_0}^{T-1}$ are analyzed in Lemma 3 in a one-step-ahead context. For our purposes, it is sufficient for them to remain bounded to accommodate some weak dependence in $\{u_{t+h}\}$.

Parts (i)–(iii) above are enough for *A*–*C* of Lemma 3. However, the MDS assumption legitimizes the use of (2.25) as the estimator of ϕ^2 that enters *D*. The full analysis of HAC-type estimators is beyond the scope of this study. However, under serial dependence, and because $\widehat{u}_{2,t+h}^2 = \widetilde{u}_{2,t+h}^2 + o_p(1)$ under our conditions, we conjecture that the HAC estimator of ϕ^2 as proposed by both Pitarakis (2023) and Pitarakis (2025) will remain consistent (see also the discussion by Fosten (2016) or Su et al. (2025), where HAC is applied after the first-step factor estimation).

2.2.2. Heterogeneous weak loadings

In the previous section, we imposed that the loadings are weaker to the same degree, or homogeneously. Alternatively, we can allow for heterogeneously weak loadings by employing the normalizing matrix $\mathbf{B}_N = \text{diag}(N^{\alpha_1/2}, \dots, N^{\alpha_r/2})$, where $1 \geq \alpha_1 > \alpha_2 > \cdots > \alpha_r > 0$, and where the weakest loading cannot still be absolutely uninformative. This means that as $N \to \infty$, some loadings are too small or too sparse for the corresponding eigenvalues to diverge at rate N, while others are relatively stronger or right-out strong such that the corresponding eigenvalues diverge at a slightly sub-linear rate or a linear rate in N. Clearly, the heterogeneous case nests the homogeneous one when $\alpha_1 = \cdots = \alpha_r$. Note that $\|\mathbf{B}_N\| \leq MN^{\alpha_1/2}$ and $\|\mathbf{B}_N^{-1}\| \leq mN^{-\alpha_r/2}$ for some positive constants M and m, which means that the order of (the inverse of) this normalization matrix is dominated by the (weakest) strongest factor loading. Again, recall that $\widehat{\mathbf{F}}_t \mathbf{D}_{Nt,r}^2 = \frac{1}{Nt}\mathbf{X}_t\mathbf{X}_t'\widehat{\mathbf{F}}_t$ by the eigenvalue–eigenvector relationship. Then, by using the fact that both $\mathbf{B}_N$ and $\mathbf{D}_{Nt,r}^2$ are diagonal, we obtain $\widehat{\mathbf{F}}_t \mathbf{D}_{Nt,r}^2 \mathbf{B}_N^{-1} = \widehat{\mathbf{F}}_t \mathbf{B}_N^{-1} \mathbf{D}_{Nt,r}^2 = \widehat{\mathbf{F}}_t \mathbf{B}_N (\mathbf{B}_N^{-2} \mathbf{D}_{Nt,r}^2) = \frac{1}{Nt}\mathbf{X}_t\mathbf{X}_t'\widehat{\mathbf{F}}_t\mathbf{B}_N^{-1}$.

As for the above, we are after an expansion of $\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}_{Nt,r}'$.[9] Hence, we use the same definitions of the scalar quantities in (2.15)–(2.17). However, we replace α with α_r (in the interim, the absence of a subscript α (α_r) means that there is no scaling in terms of N). Next, to accommodate heterogeneity in loading weakness, we redefine

$$\eta_{l,s,\alpha_r}^D = \frac{\sqrt{N}}{N^{\alpha_r}} \sum_{i=1}^{N} (N^{-1/2}\mathbf{B}_N\mathbf{f}_l)'\mathbf{B}_N^{-1}\boldsymbol{\lambda}_i e_{i,s} = \frac{\sqrt{N}}{N^{\alpha_r}}\eta_{l,s}^D, \quad (2.26)$$

$$\nu_{l,s,\alpha_r}^D = \frac{\sqrt{N}}{N^{\alpha_r}} \sum_{i=1}^{N} (N^{-1/2}\mathbf{B}_N\mathbf{f}_s)'\mathbf{B}_N^{-1}\boldsymbol{\lambda}_i e_{i,l} = \frac{\sqrt{N}}{N^{\alpha_r}}\nu_{l,s}^D, \quad (2.27)$$

where $\|N^{-1/2}\mathbf{B}_N\| = O(N^{(\alpha_1-1)/2})$, that is $O(1)$ when $\alpha_1 = 1$. Additionally, we introduce $\mathbf{Q}_{N,\alpha_r} = \mathbf{B}_N^{-1}\sqrt{N^{\alpha_r}}$, which is $O(1)$ as well. Then, by again using the commutative product, we get

$$\widehat{\mathbf{f}}_s - \mathbf{H}_{Nt,r}\mathbf{f}_s \quad (2.28)$$
$$= (N\mathbf{B}_N^{-2}\mathbf{D}_{Nt,r}^2)^{-1}\mathbf{Q}_{N,\alpha_r}^2 \left(\frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\gamma_{l,s,\alpha_r} + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\xi_{l,s,\alpha_r} + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\eta_{l,s,\alpha_r}^D + \frac{1}{t}\sum_{l=1}^{t}\widehat{\mathbf{f}}_l\nu_{l,s,\alpha_r}^D \right).$$

To handle heterogeneous weak loadings, we require adaptations of the former assumptions A.3 and A.5. More specifically,

A.3* (Heterogeneously weak loadings)

(i) $\left\|\mathbf{B}_N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1} - \boldsymbol{\Sigma}_{\boldsymbol{\Lambda}}\right\| = O_p(N^{-\alpha_r/2})$ for a positive definite $\boldsymbol{\Sigma}_{\boldsymbol{\Lambda}}$ as $N \to \infty$. The matrix $\mathbf{B}_N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1}$ is diagonal and positive definite with distinct eigenvalues ordered in a decreasing fashion.

(ii) $\mathbb{E}\left[\left(\sum_{i=1}^{N}(N^{-1/2}\mathbf{B}_N\mathbf{f}_s)'\mathbf{B}_N^{-1}\boldsymbol{\lambda}_i e_{i,t}\right)^2\right] < M$ and $\left\|\frac{1}{\sqrt{N^{\alpha_1}T}}\mathbf{F}_t'\mathbf{E}_t\boldsymbol{\Lambda}\right\| = O_p(1)$ for all t, s.

(iii) $\mathbb{E}\left(\left\|\mathbf{B}_N^{-1}\sum_{i=1}^{N}\boldsymbol{\lambda}_i' e_{i,t}\right\|^8\right) < M$ for all t.

(iv) $\left\|T^{-1}\mathbf{e}_i'\mathbf{E}_t\boldsymbol{\Lambda}\mathbf{B}_N^{-1}\right\| = O_p(N^{-\alpha_r}) + O_p\left(\sqrt{\frac{N^{\alpha_1}}{N^{\alpha_r}}}\frac{1}{\sqrt{T}}\right)$ for all $i = 1, \dots, N$ and $\forall t$.

A.5* (N, T expansion rates for heterogeneously weak loadings)

(i) $\frac{N}{N^{\alpha_r}}\frac{1}{T^{1/4}} \to c > 0$, as $(N,T) \to \infty$, (ii) $\sqrt{T}N^{-\alpha_r} \to 0$, for $\alpha_r \in (0.5, 1)$.

Finally, Lemma 4, whose proofs are in the Online Supplement, Sections 3.2.2 & 4.2, links the results of Lemmas 1, 2 and 3 to the heterogeneous loadings context. With this, the results in Theorem 1 follow directly.

Lemma 4. *Under A.1, A.2, A.3*, A.4, A.5*, and with heterogeneous loadings ($1 \geq \alpha_1 \geq \cdots \geq \alpha_r > 1/2$), as $(N,T) \to \infty$, the results of Lemmas 1, 2, 3, and Proposition 2 continue to hold with α_r in place of α.*

[9] To be precise, we are after the expansion of $(\widehat{\mathbf{F}}_t\mathbf{B}_N - \mathbf{F}_t\mathbf{B}_N\overline{\mathbf{H}}_{Nt,r}')\mathbf{B}_N^{-1}$, for $\overline{\mathbf{H}}_{Nt,r} := (N\mathbf{B}_N^{-2}\mathbf{D}_{Nt,r}^2)^{-1}\mathbf{B}_N^{-1}t^{-1}\widehat{\mathbf{F}}_t'\mathbf{F}_t\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1}$. However, as we explain in the Online Supplement, the component $\mathbf{B}_N^{-1}t^{-1}\widehat{\mathbf{F}}_t'\mathbf{F}_t\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1}$ is bounded in probability for $t = T$, as argued by Bai and Ng (2023), and we also show that it is uniformly bounded in Lemma 1 of the mathematical Online Supplement. Furthermore, using the fact that the product of $\mathbf{B}_N$ and $\mathbf{D}_{Nt,r}^2$ commutes, we can show how $\mathbf{B}_N^{-1}\overline{\mathbf{H}}_{Nt,r}\mathbf{B}_N = \mathbf{H}_{Nt,r}$.

Proof. Online Supplement, Sections 3.2 and 4.2 as counterparts to the homogeneous case.

Remark 2. It is possible to improve the rate of convergence of the in-sample estimated factors in Lemma 1 (ii.). As Lemma 8 in the Online Supplement reveals, we have for the homogeneous and heterogeneous cases, respectively,

$$(a) \quad \sup_{k_0\le t\le T-1} \frac{1}{t}\left\|\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}'_{Nt,r}\right\|^2 = O_p\left(\frac{N^2}{N^{2\alpha}}\frac{1}{T^2}\right) + O_p(N^{-\alpha}),$$

$$(b) \quad \sup_{k_0\le t\le T-1} \frac{1}{t}\left\|\widehat{\mathbf{F}}_t - \mathbf{F}_t\mathbf{H}'_{Nt,r}\right\|^2 = O_p\left(\frac{N^2}{N^{2\alpha_r}}\frac{1}{T^2}\right) + O_p(N^{-\alpha_r})$$

which does not immediately restrict α (α_r). Consequently, the rates in Lemma 2 are improved as well. This can be achieved by following (Bai & Ng, 2023) and employing a high-level assumption in A.4(vii). In this case, we do not need to analyze the sums of (2.18) and employ summability conditions on the idiosyncratics (the usual approach, as in e.g. Bai & Ng, 2002). This technique is unavoidable in Lemma 1(i) due to the out-of-sample design. For (ii), instead, we can focus on the norm of the whole time stack, and this gives the same rates as in Proposition 1 of Bai and Ng (2023). Nevertheless, the out-of-sample rates – part (i) of Lemma 1 – dictate the behavior of *A–D* of Lemma 3. Therefore, $\alpha > 0.5$ cannot be dispensed with.

Importantly, this improvement could not be achieved if the assumptions of Uematsu and Yamagata (2022) were applied to the usual principal components. This occurs due to their unspecified restrictions between $\mathbf{F}_t$ and $\mathbf{H}_{Nt,r}$ that strongly moderate the relationship between α_1 and α_r (see also footnote 2 in Bai & Ng, 2023).

3. Monte Carlo summary

We design a DGP similar to DGP2 in Pitarakis (2025), but where the factors are specified in the same way as DGP2 of Bai and Ng (2023) with strong/weak homogeneous/heterogeneous loadings. Throughout, we set the number of factors $r = 3$.

$$y_{t+1} = c + \theta_1 y_t + \boldsymbol{\beta}'\mathbf{f}_t + u_{t+1}, \quad \mathbf{f}_t \in \mathbb{R}^{r=3},$$

$$u_{t+1} = \begin{cases} NID(0,1) \ \ (Baseline), \\ \sigma_{t+1}\varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim NID(0,1), \\ \quad \sigma^2_{t+1} = \omega + \alpha u_t^2 + \eta\sigma_t^2, \quad \omega, \alpha = 0.1, \quad \eta = 0.2, \end{cases}$$

$$x_{i,t} = \boldsymbol{\lambda}_i'\mathbf{f}_t + e_{i,t},$$

$$e_{i,t} = \rho_i e_{i,t-1} + \sqrt{1-\rho_i^2}\,v_{i,t},$$

$$v_{i,t} = \begin{cases} NID(0,1) \ \ (Baseline), \\ \epsilon_{i,t} + \sum_{k=i+1}^{K}\xi(\epsilon_{i-k,t} + \epsilon_{i+k,t}), \quad \epsilon_{i,0} = 0, \\ \quad K = 5, \quad \xi = 0.4, \end{cases}$$

$$\mathbf{f}_t \sim N(0, \mathbf{I}_3), \quad \boldsymbol{\lambda}_i \sim \mathbf{G}_i\mathbf{D}\mathbf{B}_N/\sqrt{N} + \mathbf{G}_i\pi/\sqrt{N},$$
$$\mathbf{F}'\mathbf{F}/T \approx \mathbf{I}_3, \quad \mathbf{B}_N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1} \approx \mathbf{D}^2,$$

where $\mathbf{G}_i \sim N(0, \mathbf{I}_3)$, $\mathbf{D}^2 = \mathrm{diag}(3\ \ 2\ \ 1)$; $\mathbf{B}_N = \mathrm{diag}\,(N^{\alpha_1/2}\ N^{\alpha_2/2}\ N^{\alpha_3/2})$, $(\alpha_1, \alpha_2, \alpha_3) = (1, 1, 1)$ for strong homogeneous loadings, $(0.51, 0.51, 0.51)$ for weak homogeneous loadings, and $(0.51, 0.7, 1)$ for mixed strong/weak heterogeneous loadings.[10] The cross-sectional and time dimensions are $(N, T) = (800, 500)$. For the practical implementations, following Pitarakis (2025), we set $c = 1.25$, $\theta_1 = 0.5$, $\rho_i = 0.3 + N(0,1)_i \times 0.5$, $\boldsymbol{\beta} = (0,0,0)'$, and $\pi = 24$ for size; and $\boldsymbol{\beta} = (j,j,j)'$ for $j \in \{0.1, 0.2, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6\}$ for power. Before we go to the results, let us mention how we performed the same simulations using the Bai and Ng (2002) criterion, IC_{p1} (see their Eq. 9), to actively select the number of factors. We select this number once during the in-sample period to reflect the assumption that r is fixed over time. Since there is no recursion in the selection of the number of factors, this choice is justified by the results in Bai and Ng (2023), who find that in order to estimate factors with weakly convergent loadings ($\alpha > 0$), the criteria in Bai and Ng (2002) remain valid. This turned out to be identical to the results presented here, as IC_{p1} always correctly estimates the number of factors.[11]

3.1. Baseline results

We here report a summary of the Monte Carlo in the form of power curves for the different test statistics considered. The setting is the baseline, meaning that u_{t+1} is uncorrelated over time and $e_{i,t}$ is uncorrelated cross-sectionally. In Figs. 1–4, we find that both the encompassing and forecast accuracy tests to display satisfactory sizes and powers when factors are included in the alternative forecasting model. In terms of the choice of the parameters $\mu_0, \tau_0, \lambda_1^0, \lambda_2^0$, we here report the best-performing size-wise (i.e. the closest to nominal level 5%; see Online Supplement, Section 5 for the extended results). As expected, weaker loadings have a dampening effect on power, especially, of course, when the signal ($\boldsymbol{\beta}$ value) is low. This effect is more pronounced when looking at the forecast accuracy tests $g_{\widehat{f},2}$, $g^{adj}_{\widehat{f},2}$, $g_{\widehat{f},3}$, $g^{adj}_{\widehat{f},3}$, $g_{\widehat{f},4}$, and $g^{adj}_{\widehat{f},4}$, which are already affected – power-wise – by the data loss due to the sample overlapping discussed above. The power adjustment for $g_{\widehat{f},2}, g_{\widehat{f},3}, g_{\widehat{f},4}$ is paramount, because the unadjusted versions can lead to severely undersized tests, as seen in the tables in the Online Supplement, Section 5. The $g^{adj}_{\widehat{f},4}$, which averages over the second coordinate to let the MSE of the unrestricted model accumulate, has entirely analogous behavior as $g^{adj}_{\widehat{f},3}$ with some slightly higher power for lower $\boldsymbol{\beta}$. Overall, we show how all the tests have good finite sample performances when factors are included in the alternative model specification, and loadings can be either strong or weak, homogeneous or heterogeneous. We refer the reader to the Online Supplement, Section 5 for all the results, including the heterogeneous ones.

[10] Note that $\boldsymbol{\lambda}_i$ is simulated differently from in Bai and Ng (2023). The reason is that the rate of $\|\mathbf{B}_N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1} - \boldsymbol{\Sigma}_{\boldsymbol{\Lambda}}\|$ plays an important role in our asymptotic analysis, whereas it did not matter for Bai and Ng (2023). This simulation method mimics A.3(i), because we can show that under such a design, $\mathbf{B}_N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda}\mathbf{B}_N^{-1} = \mathbf{D}^2 + O_p(N^{-\alpha_r/2})$ as desired.

[11] We shall mention, however, that others of the Bai and Ng (2002) criteria did not perform as well as IC_{p1} in selecting the factors; we report IC_{p1}, as it is the most remarkable.

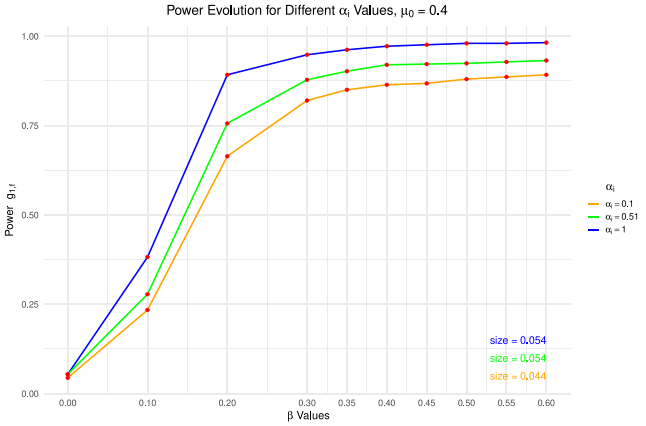


Fig. 1. $g_{\widehat{f},1}$, encompassing.

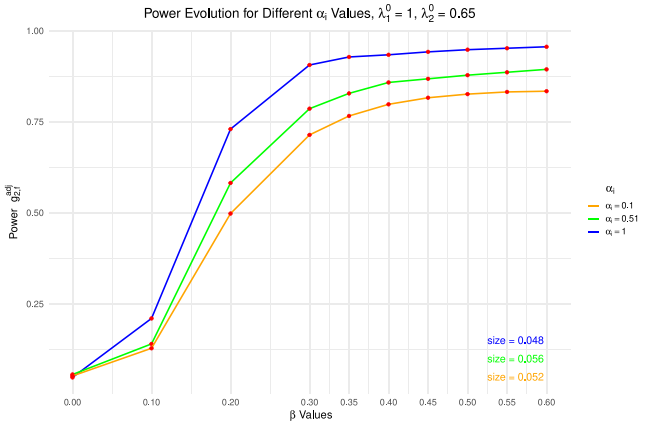


Fig. 2. $g^{adj}_{\widehat{f},2}$, forecast accuracy.

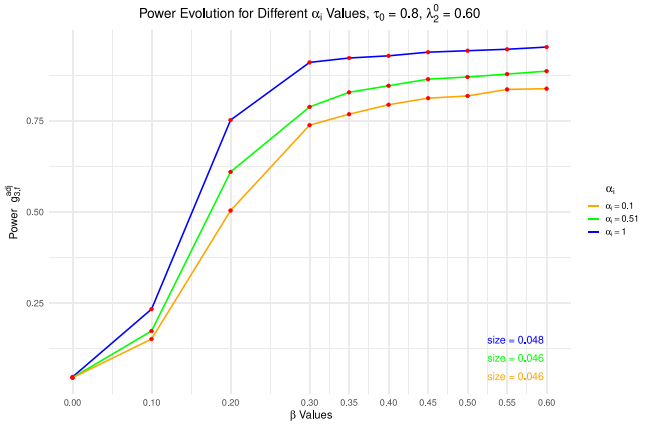


Fig. 3. $g^{adj}_{\widehat{f},3}$, forecast accuracy.

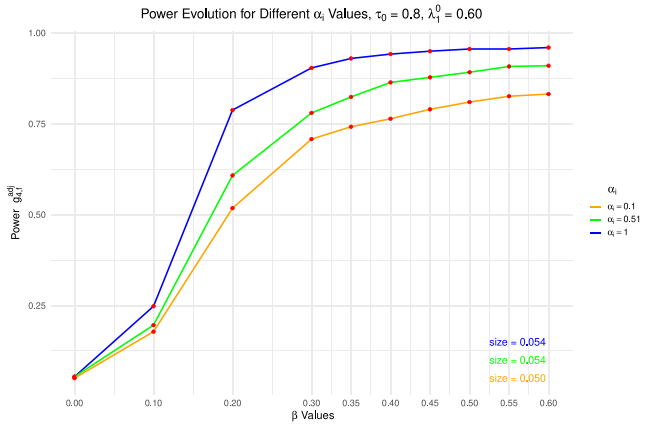


Fig. 4. $g^{adj}_{\widehat{f},4}$, forecast accuracy.

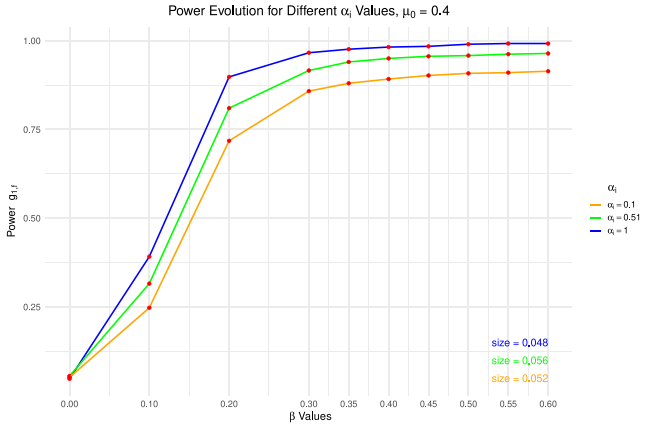


Fig. 5. $g_{\widehat{f},1}$, encompassing.

3.2. Cross-section dependence

In Figs. 5–8, we introduce weak cross-section dependence in $e_{i,t}$ similarly to Stauskas and Westerlund (2022). We see that the results are virtually the same as in the baseline scenario. This signals robustness to both temporal and, for example, spatial dependence structures in idiosyncratics.

3.3. Conditional heteroskedasticity

In Figs. 9–12, we introduce, on top of the cross-section dependence of the idiosyncratics, conditional heteroskedasticity in the forecast errors u_{t+1}, in the form of a GARCH(1, 1), as specified above. Throughout the simulations, we still use (2.25) to estimate ϕ^2. The main effect for all tests is a mild inflation of the size,[12] which is however milder for $g^{adj}_{\widehat{f},4}$ compared to $g_{\widehat{f},1}$, $g^{adj}_{\widehat{f},2}$, and $g^{adj}_{\widehat{f},3}$. The power is instead higher for lower values of the coefficient $\boldsymbol{\beta}$ (e.g. $\boldsymbol{\beta} = [0.10, \ldots, 0.30]$) if compared to the previous results, though this might just be a byproduct of the size increase or of the GARCH time-varying variance structure, which potentially makes certain periods in the sample more informative. At the same time, even though (2.25) estimates unconditional variance, it can be sensitive to conditional heteroskedasticity, especially as it utilizes 4th moments. While a theoretical justification for HAC-type corrections in our testing framework is left for future research, we experimented with the practical implementations proposed by Pitarakis (2023) for $g_{\widehat{f},1}$, specifically using Newey–West and Andrews standard errors. These methods demonstrate good empirical performance in our simulations and are thus recommended for practitioners, particularly in settings where conditional heteroskedasticity – such as GARCH-type volatility – is likely present in the forecast errors. The Online Supplement contains heterogeneous weakness results as well.

Remark 3. The large combinations of (N, T) in our experiments are selected to demonstrate the asymptotic properties of the tests and the effect of weaker loadings. In Section 5.2 of the Online Supplement, we provide a full set of simulations for smaller samples (e.g. $N = 100$,

[12] As one would expect, if the persistence of the past conditional variance of u_t is high(er), i.e. the η of the GARCH(1, 1) is large(r), the size would suffer more. In such cases HAC-type corrections of the variance are recommended.

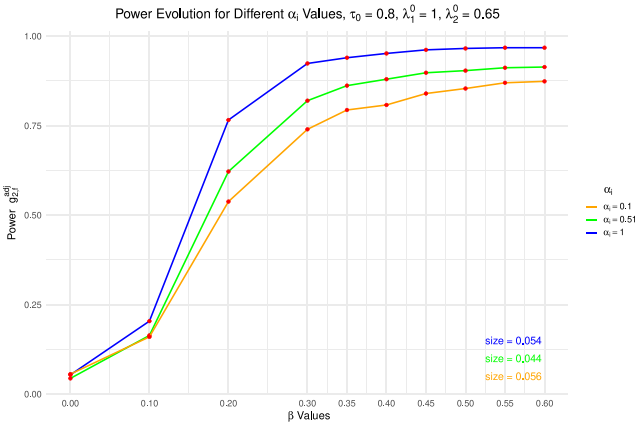


Fig. 6. $g_{\widehat{f},2}^{adj}$, forecast accuracy.

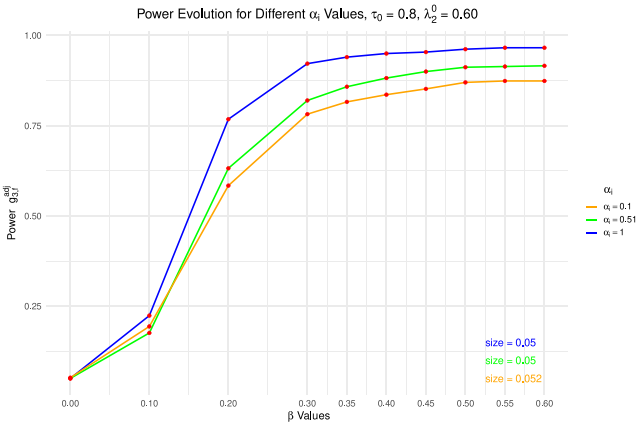


Fig. 7. $g_{\widehat{f},3}^{adj}$, forecast accuracy.

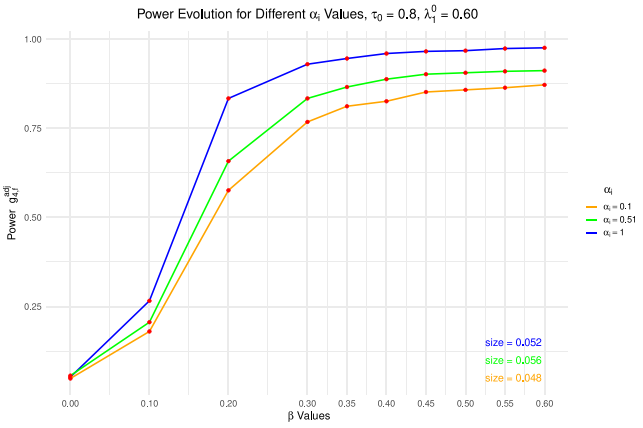


Fig. 8. $g_{\widehat{f},4}^{adj}$, forecast accuracy.

$T = 200, 350$) to reflect more practical scenarios. While natural, the loss in power is not substantial. Nevertheless, small sample performance can be improved by bootstrap, because Assumptions 1–5 in Gonçalves and Perron (2014) are analogous to ours. Also, their requirement $\sqrt{T}N^{-1} \to c$ for $0 \leq c < \infty$ is analogous to, for $c = 0$, our current Assumption A.5, where $\sqrt{T}N^{-\alpha} \to 0$ for $\alpha \in (0.5, 1)$. As such, their best-case (i.e. without any asymptotic bias term) bootstrap can be used in our context too. The bootstrap DGP in this case can be constructed

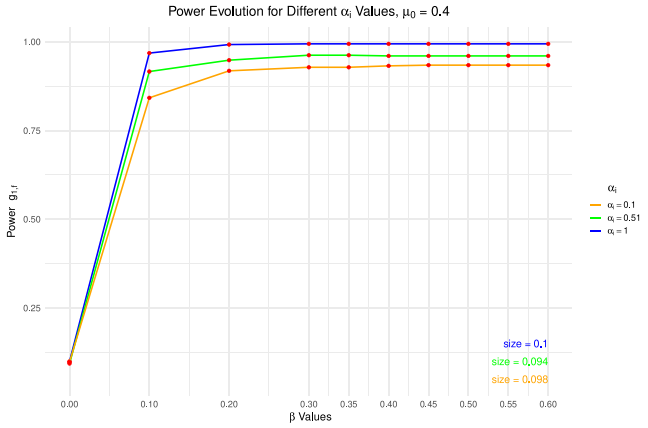


Fig. 9. $g_{\widehat{f},1}$, encompassing.

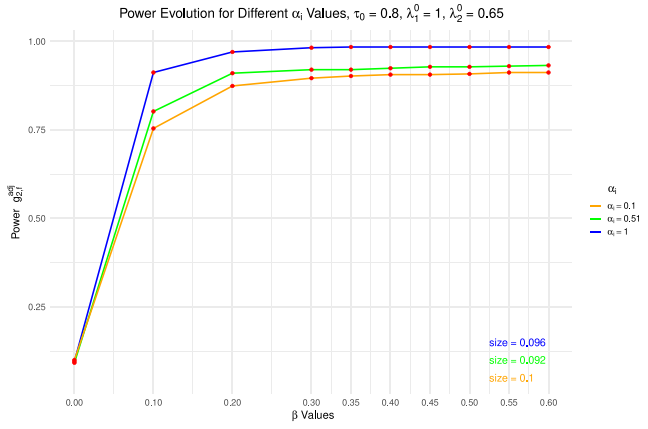


Fig. 10. $g_{\widehat{f},2}^{adj}$, forecast accuracy.

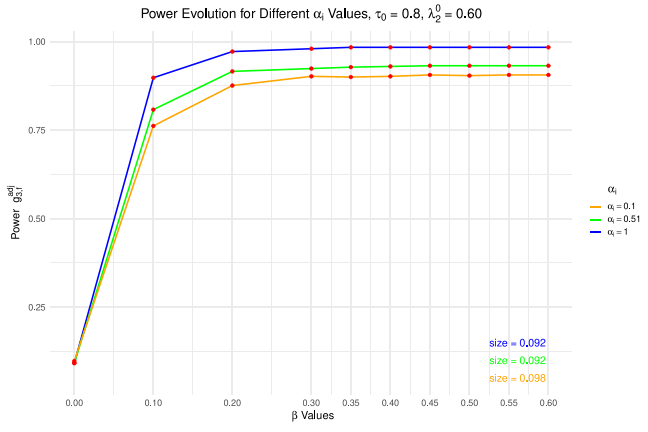


Fig. 11. $g_{\widehat{f},3}^{adj}$, forecast accuracy.

using the wild bootstrap scheme described in Section 4 of their study. Specifically, before Algorithm 1 starts, we generate $\mathbf{X}_t^* = \widehat{\boldsymbol{\Lambda}}\widehat{\mathbf{f}}_t + \mathbf{e}_t^*$ and $y_{t+1}^* = \widehat{\boldsymbol{\delta}}'\widehat{\mathbf{z}}_t + u_{t+1}^*$, where $\widehat{\mathbf{z}}_t = (\mathbf{w}_t', \widehat{\mathbf{f}}_t')'$, while $\widehat{\mathbf{f}}_t$, $\widehat{\boldsymbol{\Lambda}}$ and $\widehat{\boldsymbol{\delta}} = (\widehat{\boldsymbol{\theta}}', \widehat{\boldsymbol{\beta}}')'$ are the initial estimates obtained using the whole time series sample $t = 1, \ldots, T$. Next, $\mathbf{e}_t^* = (\nu_{1,t}\widehat{e}_{1,t}, \ldots, \nu_{N,t}\widehat{e}_{N,t})'$ and

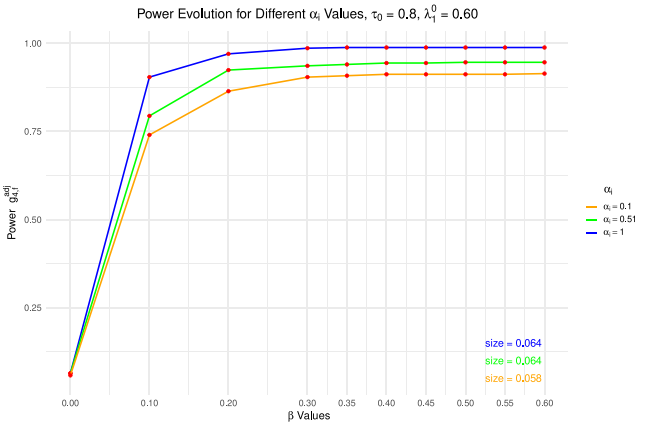


Fig. 12. $g_{\hat{f},4}^{adj}$, forecast accuracy.

$u_{t+1}^* = \eta_{t+1}\widehat{u}_{t+1}$, where $\nu_{i,t}$ and η_t are IID(0, 1) mutually independent variables that scale the residuals from the initial regression. Eventually, Algorithm 1 is implemented for $b = 1, \ldots, B$ bootstrap samples generated according to the wild scheme.

4. Empirical applications

4.1. Inflation forecasting

We partially replicate the inflation forecast exercise conducted by Pitarakis (2023). Our focus here is to explore whether *global* inflation can enhance the accuracy of country-level inflation forecasts. This debate is not new; previous studies have provided evidence that global inflation trends can significantly improve domestic inflation forecasts (see among others Ciccarelli & Mojon, 2010; Monacelli & Sala, 2009). However, other research suggests that the relevance of global inflation in forecasting domestic rates may stem solely from its ability to capture slow-moving trends in inflation (Mikolajun & Lodge, 2016), or that a global inflation factor improves forecasting accuracy primarily at longer horizons (Gillitzer & McCarthy, 2019). The question is, also, how to measure global inflation. One approach is to calculate a grand average of country-level inflation rates, represented as $\bar{\pi}_t = N^{-1}\sum_{i=1}^{N}\pi_{i,t}$. Another method involves treating global inflation as a few latent factors that can be estimated using principal components from the pool of country-level inflation rates. The former approach is employed by Pitarakis (2023) to illustrate his encompassing test, while we will utilize the latter method to demonstrate how JYP's encompassing tests work in an empirically relevant context where principal component factors are considered. There are several reasons why a PCA of global inflation serves as a more accurate measure than a simple grand average of inflation rates. A straightforward sample mean does not consider differences in economic size, inflation volatility, or other factors that may make some countries' inflation rates more indicative of global trends than others. In contrast, principal components can uncover patterns of co-movement in inflation rates that might not be evident from the raw data. Additionally, it is more robust to potential outliers and can adapt to time-varying relationships among countries' inflation rates. However, one could argue that if we assume the existence of only one factor, then all the information contained in that factor is effectively the same as a cross-sectional average of the countries' inflation rates. This would be true, upon essentially three assumptions: (i) the existence of an exact factor model underlying the data, i.e. $\pi_{i,t} = \boldsymbol{\lambda}_i'\mathbf{f}_t + e_{i,t}$ with $Cov(e_{i,t}, e_{j,s}) = 0$, $t, s \in \mathbb{Z}$, $i, j = 1, \ldots, N$, $i \neq j$; (ii) a large cross-sectional dimension N (in principle, $N \to \infty$); (iii) all or most of the loadings being non-zero (i.e. the pervasiveness of factors). Assume that the loadings are fixed. If all these are satisfied, it is clear how $Var(\bar{\pi}_t) = N^{-2}(\sum_i \lambda_i)^2 Var(f_t) + N^{-2}Var(\sum_i e_{i,t}) \to \bar{\lambda}^2 = Var(\bar{\chi}_t)$, as $N \to \infty$, meaning how the aggregation of the observed data recovers the same information contained in the factor. Now, (i) is clearly too strong (see also our A.4(v)), (ii) is what is referred to as the 'blessing of dimensionality' but in practice it clearly depends on the available data, and (iii) is precisely what is challenged by the weaker loadings treated in Section 2. Hence, there are good reasons to re-run this exercise using factors and employing JYP's tests to check whether global inflation computed by means of principal component factors improves the country-level inflation forecast. We employ the same dataset provided in Pitarakis (2023),[13] based on the World Bank's global inflation database and covering the period from 1970–2023 for 23 countries at quarterly frequency. We choose the tuning parameters in line with the simulations to balance the power and size.[14]

The base-line model is an $AR(1)$, while the alternative model is a factor-augmented $AR(1)$, where factors are recursively estimated and their number is determined via the Bai and Ng (2002) information criterion (IC_{p1}, max number $= 10$).[15] In Table 4.1, we find patterns of significance across the four test statistics, though rarely

[13] Freely available on JYP's GitHub page: https://github.com/jpitarakis/Multi-Step_Encompassing.

[14] For the exact treatment and transformations of the raw data, see Pitarakis (2023), Section 7.

Table 4.1
p-values: One quarter ahead.

Country	*p*-value $g_{\widehat{f},1}$	*p*-value $g^{adj}_{\widehat{f},2}$	*p*-value $g^{adj}_{\widehat{f},3}$	*p*-value $g^{adj}_{\widehat{f},4}$
United States of America (USA)	**0.007**	0.469	**0.008**	0.983
United Kingdom (GBR)	0.408	0.767	0.742	**0.003**
Japan (JPN)	0.284	0.388	0.094	**0.007**
France (FRA)	**0.003**	**0.023**	**0.010**	0.877
Germany (DEU)	0.863	0.186	0.539	0.540
Spain (ESP)	**0.000**	**0.000**	**0.000**	0.936
Italy (ITA)	0.859	0.306	0.845	0.392
Netherlands (NLD)	**0.001**	**0.000**	**0.002**	0.192
Luxembourg (LUX)	**0.000**	0.194	**0.044**	0.372
Canada (CAN)	0.070	0.711	0.142	0.736
Ireland (IRL)	0.196	0.574	0.067	0.110
Finland (FIN)	0.898	0.972	0.956	0.121
New Zealand (NZL)	0.998	0.971	0.969	0.336
Greece (GRC)	**0.004**	0.359	0.525	**0.013**
Portugal (PRT)	0.241	0.355	0.329	**0.000**
Norway (NOR)	0.257	0.905	0.866	0.168
South Korea (KOR)	0.459	0.714	0.569	**0.004**
Denmark (DNK)	0.414	0.473	0.527	**0.000**
Sweden (SWE)	0.496	0.855	0.787	**0.000**
Australia (AUS)	0.635	0.793	0.869	**0.022**
Austria (AUT)	**0.002**	**0.003**	**0.009**	**0.000**
Belgium (BEL)	0.233	0.173	0.485	0.348
Switzerland (CHE)	**0.000**	0.138	**0.000**	0.312

Notes: *AR*(1) vs. factor-augmented *AR*(1); number of factors selected with Bai and Ng (2002) IC_{p1}, $\mu_0 = 0.40$, and $\tau_0 = 0.8$ for $g^{adj}_{\widehat{f},2}$: $\lambda^0_1 = 1, \lambda^0_2 = 0.65$; for $g^{adj}_{\widehat{f},3}$: $\lambda^0_2 = 0.6$; for $g^{adj}_{\widehat{f},4}$: $\lambda^0_1 = 0.6$.

all of them at once (only AUT). $g^{adj}_{\widehat{f},4}$ finds significant better forecast accuracy one quarter ahead when using a PCA of global inflation in nine of the 23 countries. It is followed by the encompassing test $g_{\widehat{f},1}$ with eight countries (nine with CAN if considering a 10% nominal level), $g^{adj}_{\widehat{f},3}$ with seven and $g^{adj}_{\widehat{f},2}$ with four. Noticeably, for some large economies such as the USA, FRA, and ESP, at least two test statistics of the four are found to be significant (JPN too, if considering a 10% nominal level). Some other large/medium-to-large economies like GBR, DEU, and ITA have only one or no significance at all. Overall, this paints a mixed picture with regard to the use of global inflation to better forecast country-level inflation, which clearly seems to be a country-specific issue. These tests can therefore be used to gain a sense as to whether a PCA of global inflation might help beyond the simple AR specification. In the Online Supplement, Section 6, we repeat the exercise without factors but using a grand average of country-level inflations (which excludes the country of reference each time) as global inflation. What comes out is an overly suspicious abundant significance across most countries and all tests. Outliers and multicollinearity are likely to affect these results, as evident from the boxplots and pairwise correlation heatmap in Fig. 13 and Fig. 14, respectively. It is therefore safe to say that the analysis, including the PCA of global inflation, is much more trustworthy.

Remark 4. We suggest that the potential macroeconomics practitioners interpret the outcomes of the four tests as some sort of battery of checks to be used jointly in understanding whether the factor augmentation is worth it to improve the quality of the forecast. As outlined above, acknowledging that each test has its own nuance, $g_{\widehat{f},1}$ should be given more weight in the decision. This is because, as explained above, due the overlapping evaluation over the effective sample size for the case of $g^{adj}_{\widehat{f},2}$, $g^{adj}_{\widehat{f},3}$, and $g^{adj}_{\widehat{f},4}$, some data are lost in the MSE comparison, which does not occur for $g_{\widehat{f},1}$. As such, a joint significance of $g_{\widehat{f},1}$ and at least one of $g^{adj}_{\widehat{f},2}$, $g^{adj}_{\widehat{f},3}$, and $g^{adj}_{\widehat{f},4}$ should possibly be taken as a solid indication that considering a factor-augmented specification can increase the forecast accuracy (see Table 4.1).

4.2. FRED-MD forecasting

We continue the empirical exercise by considering more macroeconomic series, this time from an even larger pool of series. Table 4.2 collects the *p*-values for the tests of encompassing and forecast accuracy described in Section 2, using the same tunings as in the previous application in Section 4.1, for 14 important macroeconomic series taken from the FRED-MD dataset McCracken and Ng (2016) (July 2024 vintage, starting in January 1960). The FRED-MD dataset includes a wide range of U.S. macroeconomic series, such as output (e.g. industrial

[15] It is well known how the Bai and Ng (2002) criteria depend quite substantially on the maximum number of factors as selected by the practitioner, as well as by the relative magnitude of N, T (see e.g. Forni et al., 2009). In this case, we find on average that all criteria, including IC_{p1} but excluding AIC_3, return the maximum as the estimated number of factors. We experimented using AIC_3 and the ABC criterion of Alessi et al. (2010) too, which both on average estimated 4/5 common factors, but we found a completely similar picture in terms of significance, with only slightly higher *p*-values.

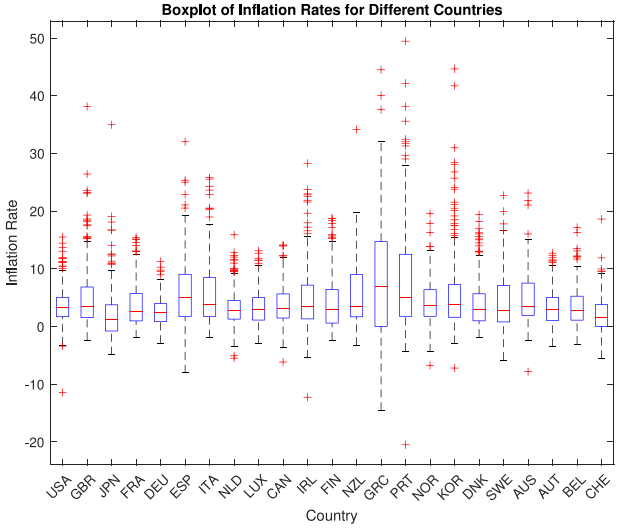


Fig. 13. Inflation boxplot.

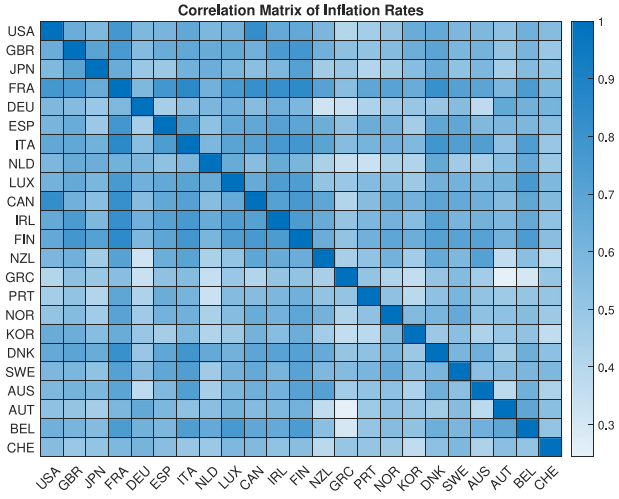


Fig. 14. Correlation heatmap.

production), income, labor market indicators (e.g. unemployment rate, payrolls), consumer and producer prices, money supply, interest rates, exchange rates, and financial variables like stock indices and spreads. These series capture economic activity, inflation dynamics, monetary policy, and financial market conditions. All series are cleaned and stationary transformed, as prescribed by the Matlab routines provided with FRED-MD. After the necessary cleaning of NAs and outliers, the final dataset contains 117 series for 772 data points. Again, we choose the tuning parameters in line with the simulations to balance the power and size (see our extensive simulations in the Online Supplement).

We forecast these 14 macroeconomic series one month ahead using the principal component factors in the alternative forecasting specification (i.e. $AR(1)$ vs. factor-augmented $AR(1)$, where factors are estimated via Bai and Ng (2002) IC_{p1} with max number= 15).[16][17] With an average of seven estimated common factors, the results are overwhelmingly significant across the four tests. Apart from 'Real personal consumption expenditures' (DPCERA3M086SBEA), all the series have at least one significant (at a nominal level of 5%) test statistic. This shows how even in a high(er) dimensional dataset such as the FRED-MD, these test statistics are very useful for assessing

16 We choose to follow a common convention in the factor modeling literature by setting a relatively high maximum number of factors, from which the optimal number is selected using IC. The goal of this is to ensure that the true number of factors is not underestimated due to an arbitrarily restrictive upper bound.

17 We collect the full names of the 14 FRED-MD macroeconomic time series in the Online Supplement (Section 6, Table 6.2).

Table 4.2
FRED-MD forecast accuracy & encompassing.

Series	p-value $g_{\widehat{f},1}$	p-value $g^{adj}_{\widehat{f},2}$	p-value $g^{adj}_{\widehat{f},3}$	p-value $g^{adj}_{\widehat{f},4}$
UNRATE	**0.000**	**0.000**	**0.000**	0.628
CPIAUCSL	**0.004**	**0.029**	**0.000**	1.000
DPCERA3M086SBEA	0.168	0.112	0.510	0.219
S&P 500	**0.030**	**0.021**	0.118	0.898
PAYEMS	**0.000**	**0.000**	**0.014**	0.566
INDPRO	**0.000**	**0.000**	**0.000**	0.919
TB3SMFFM	0.986	0.885	0.885	**0.000**
HOUST	**0.002**	**0.000**	**0.000**	0.999
M1SL	**0.000**	**0.000**	**0.000**	0.999
M2SL	**0.000**	**0.003**	**0.018**	0.796
OILPRICEx	**0.000**	**0.000**	**0.003**	0.893
GS10	0.999	0.963	0.985	**0.000**
RPI	0.066	**0.012**	**0.034**	0.836
BUSLOANS	0.117	**0.037**	0.109	0.881

Notes: One month ahead, $AR(1)$ vs. factor-augmented $AR(1)$; number of factors selected with Bai and Ng (2002) IC_{p1}, $\mu_0 = 0.40$, and $\tau_0 = 0.8$, for $g^{adj}_{\widehat{f},2}$: $\lambda^0_1 = 1$, $\lambda^0_2 = 0.65$; for $g^{adj}_{\widehat{f},3}$: $\lambda^0_2 = 0.6$; for $g^{adj}_{\widehat{f},4}$: $\lambda^0_1 = 0.6$.

whether principal component factors could be employed to improve the forecast of macroeconomic series.

5. Conclusion

We developed a theoretical framework that allows the forecast accuracy and encompassing tests proposed by Pitarakis (2023, 2025) to be applied when the alternative forecasting model incorporates estimated principal component factors. These factors can have loadings that are either strong or weak, whether homogeneously or heterogeneously distributed. Our theoretical findings were supported by both a Monte Carlo simulation and two empirical applications in macroeconomic forecasting.

CRediT authorship contribution statement

Luca Margaritella: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Ovidijus Stauskas:** Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.11.005.

Data availability

The detailed replication package can be found in this GitHub repository: GitHub - New Tests of Equal Forecast Accuracy for Factor-Augmented Regressions with Weaker Loadings.

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